Factor-Adjusted Smoothed Rank Estimation for High-Dimensional Accelerated Failure Time Models

Weineng ZHU · Yang JU

Abstract Factor adjustment separates common and feature-specific variation in high-dimensional accelerated failure time (AFT) regression. Under right censoring, the estimated factor coordinates must be connected to the original sparse slope. We develop a smoothed rank estimator with unpenalized clinical and factor effects and an ℓ_1 penalty on molecular coefficients. An exact reference representation preserves the ordinary-model predictor and separates design perturbation from additional-factor approximation. Under common-error, conditional-censoring, local-information, and factor-design conditions, finite-sample coefficient bounds require no moments of the log-time error. The analysis accommodates factors estimated from the same covariate sample. Column-scale control of pairwise fluctuations avoids an additional logarithmic loss. A full-coordinate stationarity criterion transfers the bounds to numerical candidates. In three fixed strong-dependence samples, factor adjustment has lower maximum active curvature and uses fewer gradient evaluations to reach common accuracy. One active-face Newton step completes each unadjusted fit. Training selection yields twelve fits at the prescribed tolerance and distinguishes sparse residual effects from dense factor-mediated prediction.

Keywords Accelerated failure time · Right censoring · Factor adjustment · High-dimensional estimation · Smoothed rank regression · Computational accuracy

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1 Introduction

High-dimensional survival analysis seeks interpretable associations between incomplete event times and many dependent covariates. Molecular measurements often combine a low-dimensional common component with feature-specific variation. In an accelerated failure time (AFT) model, the scientific target is a sparse slope on the observed covariates. Factor adjustment offers a representation of their dependence, but the factors used in estimation are themselves learned from the sample. A statistical analysis must therefore connect the original slope to the estimated factor coordinates while accounting for right censoring. We develop that connection for a smoothed rank estimator.

Rank-based AFT regression estimates conditional log-time associations with an unspecified error distribution (Wei, 1992; Jin et al., 2003). Regularized Gehan estimation, joint clinical adjustment, and scalable high-dimensional algorithms are developed by Cai et al. (2009), Johnson (2009), and Suder and Molstad (2022). Distribution-function and parameter-perturbation smoothing produce differentiable censored-rank criteria (Heller, 2007; Johnson and Strawderman, 2009). Related computational and robust formulations include Choi and Choi (2021), Li et al. (2021), and Seon et al. (2026, preprint, arXiv:2606.31094). Our estimator uses a common scalar Gaussian convolution of the positive-part loss, so the bandwidth enters the analysis through an explicit smoothing-bias term.

Factor adjustment retains the common component as an unpenalized regression block. FarmSelect develops this construction for regularized estimation (Fan et al., 2020); Bayle and Fan (2022, preprint, arXiv:2210.01067) study factor-augmented Cox regression. For censored rank estimation, the estimated factor representation and the patient pairs in the loss share the same covariate sample. This dependence requires a joint treatment of score centering, pairwise fluctuations, smoothing bias, and estimated-design curvature. The unpenalized clinical and factor coefficients must also be included in the error geometry.

Our primary target is $\boldsymbol{\beta}_0$ in the ordinary observed-covariate model

$$\begin{aligned} \boldsymbol{X}_i &= \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i, \\ \log T_i &= \boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i, \end{aligned} \tag{1.1}$$

where $\boldsymbol{X}_i \in \mathbb{R}^p$, $\boldsymbol{Z}_i \in \mathbb{R}^d$, and the factor dimension K is small. The predictor is equivalently $\boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{u}_i^\top \boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top \boldsymbol{\gamma}_0$, with $\boldsymbol{\gamma}_0 = \boldsymbol{B}^\top \boldsymbol{\beta}_0$. We estimate the common and residual components by principal component analysis (PCA), then minimize a smoothed Gehan loss with an ℓ_1 penalty on $\boldsymbol{\beta}$. Clinical and factor coefficients are estimated jointly and remain unpenalized. The estimator allows the factor coefficient to vary freely; it contains the ordinary fitted predictor as a special case while retaining $\boldsymbol{\beta}_0$ as its sparse target. Section 2 also considers an additional factor effect relative to a specified latent representation.

The main contribution is a finite-sample analysis with same-sample factor estimation. An exact reference representation preserves the centered ordinary-model predictor and separates estimated-design perturbation from the approximation associated with an additional factor effect. Conditional on the full design, an event-weighted residual-difference identity centers the unsmoothed score. Pairing and contraction bound the empirical contrasts at the design's column root-mean-square scale, avoiding the extra logarithmic loss of an entrywise-envelope argument. Local event information and a regularization argument that includes the unpenalized block yield coefficient bounds and finite attainment on the same statistical event. Primitive factor conditions supply the required estimated-design geometry. The analysis uses integrable loss differences and accommodates log-time errors without moments under the stated density and censoring conditions.

A second contribution connects statistical and numerical accuracy. A complete stationarity residual transfers the coefficient guarantee to finite numerical candidates on the same fixed-tuning event. Event-row blocking evaluates the full pairwise gradient efficiently. We provide a reference proximal-gradient convergence result and implement the same objective using accelerated iterations with qualified active-face Newton refinement. These components separate the statistical target, the search algorithm, and the accuracy required of its output.

The numerical illustrations connect this framework to computational geometry and training selection. In three fixed strong-dependence samples, factor adjustment has lower maximum active curvature and reaches common stationarity thresholds with fewer gradient evaluations. Residual projections identify weak-curvature directions in the late iterates; one Newton step completes each of the three unadjusted fits on its existing active face. A shared fold-specific selection procedure produces twelve full-training refits at the prescribed tolerance. The selected models also distinguish sparse residual effects from the dense molecular slope induced by the common factors. Each illustration addresses the numerical or representational role of the estimator; independent repeated-sample risk and biomedical applications are separate evaluation targets.

Section 2 defines the observation scheme and regression targets. Section 3 constructs the estimator and frozen prediction map. Sections 4 and 5 develop the statistical and numerical results. Section 6 presents the computational and training-selection illustrations, and Section 7 discusses their implications. Online Resource 1 contains the proofs in Supplementary Appendices A–E and the reproducibility and implementation details in Appendices F–G.

2 Model and estimation targets

2.1 Observed data

Bold letters denote vectors and matrices; indexed scalar entries are italic. For independent patients, we observe

$$\begin{aligned} O_i &= (Y_i, \Delta_i, \boldsymbol{X}_i, \boldsymbol{Z}_i), \\ Y_i &= \log\{\min(T_i, C_i)\}, \\ \Delta_i &= \mathbf{1}\{T_i \le C_i\}, \\ i &= 1, \dots, n, \end{aligned} \tag{2.1}$$

where $T_i \in (0, \infty)$ is the event time, $C_i \in (0, \infty]$ is the censoring time, $\boldsymbol{X}_i \in \mathbb{R}^p$ contains baseline molecular measurements, and $\boldsymbol{Z}_i \in \mathbb{R}^d$ contains prespecified clinical covariates. We consider a single right-censored endpoint observed from time zero, with fully observed baseline covariates and Y_i on the natural-logarithmic scale.

The covariates have prespecified coordinate scales and finite second moments. We absorb population means into the unspecified error location and take $E(\boldsymbol{Z}_i) = E(\boldsymbol{X}_i) = 0$ for notation. The clinical design excludes a constant column; Section 2.6 defines sample centering. The dimension $p = p_n$ and sparsity may grow with n, whereas d and the true factor dimension K are fixed in the theory. Dependence of the joint law on n is suppressed.

2.2 Factor representation and coordinate normalization

Suppose the molecular covariates admit the representation

$$\begin{aligned} \boldsymbol{X}_i &= \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i, \\ \boldsymbol{B} \in \mathbb{R}^{p\times K}, &\quad \boldsymbol{f}_i \in \mathbb{R}^K, \quad \boldsymbol{u}_i \in \mathbb{R}^p, \end{aligned} \tag{2.2}$$

with

$$E(\boldsymbol{f}_i) = 0, \quad E(\boldsymbol{u}_i) = 0, \quad E(\boldsymbol{f}_i\boldsymbol{f}_i^\top) = \boldsymbol{I}_K, \quad E(\boldsymbol{f}_i\boldsymbol{u}_i^\top) = 0. \tag{2.3}$$

The factors and idiosyncratic components are latent. Define the full design vector

$$\boldsymbol{D}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top)^\top \in \mathbb{R}^P, \qquad q = d + K, \qquad P = p + q.$$

Equation (2.3) imposes second-moment orthogonality between factors and idiosyncratic components, while allowing dependence within and between covariate blocks. Clinical covariates may be associated with either component.

For the pervasive-factor regime, the basic structural conditions are

$$\begin{aligned} c_B\boldsymbol{I}_K \preceq p^{-1}\boldsymbol{B}^\top\boldsymbol{B} \preceq C_B\boldsymbol{I}_K, \\ \|\boldsymbol{\Sigma}_u\|_{\mathrm{op}} \le C_u, \\ \boldsymbol{\Sigma}_u = \mathrm{Cov}(\boldsymbol{u}_i), \end{aligned} \tag{2.4}$$

where the positive constants are uniform in n, p. Equation (2.4) supplies the spectral separation used in factor-adjusted regression (Fan et al., 2020); Section 4 adds distributional, loading, and sparse-design conditions for estimated-factor guarantees.

For an orthogonal $K \times K$ matrix $\boldsymbol{Q}$, the transformation

$$\boldsymbol{f}_i^* = \boldsymbol{Q}\boldsymbol{f}_i, \qquad \boldsymbol{B}^* = \boldsymbol{B}\boldsymbol{Q}^\top \tag{2.5}$$

preserves the common component and (2.3); the factor coefficients transform with the same rotation. Population rotation and sample normalization $\widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{F}}/n = \boldsymbol{I}_K$ play separate roles. Finite-sample alignment matrices can be nonorthogonal (Bai and Ng, 2013). Section 2.5 establishes slope identification for these coordinate representations.

2.3 AFT models and the sparse coefficient target

The primary model is an ordinary sparse AFT model,

$$\mathcal{M}_0 : \qquad \log T_i = \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top\boldsymbol{\beta}_0 + \varepsilon_i. \tag{2.6}$$

Factor adjustment retains $\boldsymbol{\beta}_0$ as the observed-covariate slope and represents dependence through common and residual components. We also consider the extension

$$\begin{aligned} \mathcal{M}_1 : \qquad \log T_i &= \eta_i^0 + \varepsilon_i, \\ \eta_i^0 &= \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top\boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top\boldsymbol{\xi}_0 \\ &= \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{u}_i^\top\boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top\boldsymbol{\gamma}_0, \\ \boldsymbol{\gamma}_0 &= \boldsymbol{B}^\top\boldsymbol{\beta}_0 + \boldsymbol{\xi}_0. \end{aligned} \tag{2.7}$$

Thus $\mathcal{M}_0$ is the submodel $\boldsymbol{\xi}_0 = 0$. Additional factor contributions also arise in factor-augmented hazard regression (Bayle and Fan, 2022, preprint, arXiv:2210.01067). In $\mathcal{M}_1$, the coefficient target is defined relative to the specified latent factor representation.

Write

$$\begin{aligned} \boldsymbol{a}_0 &= (\boldsymbol{\theta}_0^\top, \boldsymbol{\gamma}_0^\top)^\top, \\ \boldsymbol{\vartheta}_0 &= (\boldsymbol{a}_0^\top, \boldsymbol{\beta}_0^\top)^\top, \\ S &= \operatorname{supp}(\boldsymbol{\beta}_0), \\ s &= |S|, \\ k &= s + q. \end{aligned} \tag{2.8}$$

Sparsity is imposed on $\boldsymbol{\beta}_0$; the q clinical and factor coefficients are unpenalized. The target $\boldsymbol{\beta}_0$, its support, and η_i^0 are invariant under (2.5) when $\boldsymbol{\gamma}_0^* = \boldsymbol{Q}\boldsymbol{\gamma}_0$ and $\boldsymbol{\xi}_0^* = \boldsymbol{Q}\boldsymbol{\xi}_0$. Individual factor coordinates transform with the basis. Even for fixed K, the magnitude of $\boldsymbol{\gamma}_0 = \boldsymbol{B}^\top\boldsymbol{\beta}_0$ in $\mathcal{M}_0$ can grow with the sparse signal.

The error is independent of $\boldsymbol{D}_i$ and has a common distribution function F_ε, defining a location-shift model with unrestricted location and no mean or variance requirement. If $F_\varepsilon^{-1}(\tau)$ denotes its generalized quantile, the model gives

$$Q_{T|\boldsymbol{D}_i}(\tau) = \exp\{\eta_i^0 + F_\varepsilon^{-1}(\tau)\}, \qquad 0 < \tau < 1. \tag{2.9}$$

Within covariate support, a unit change in u_{ij} at fixed remaining coordinates multiplies conditional event-time quantiles by $\exp(\beta_{0j})$. The coefficients describe conditional log-time associations: larger η_i^0 corresponds to longer event times. The intercept is absorbed into F_ε and cancels from pairwise residual comparisons.

2.4 Censoring and observable local event information

Assumption 2.1 (Common error and conditional censoring)**.** The triples $(\boldsymbol{D}_i, \varepsilon_i, C_i)$ are independent and identically distributed, and

$$\varepsilon_i \perp \boldsymbol{D}_i, \qquad \varepsilon_i \perp C_i \mid \boldsymbol{D}_i. \tag{2.10}$$

The common error distribution is absolutely continuous, with density f_ε satisfying $f_\varepsilon(t) \le L_\varepsilon < \infty$ for almost every $t \in \mathbb{R}$.

Assumption 2.1 yields $\Pr(\varepsilon_i \le t \mid \boldsymbol{D}_i, C_i) = F_\varepsilon(t)$ and $T_i \perp C_i \mid \boldsymbol{D}_i$. Censoring may depend on any covariate block and may include atoms, such as an administrative end of follow-up. The density condition applies to the event error. Recorded-time rounding is handled by a separately specified observation or tie convention.

In $\mathcal{M}_0$, η_i^0 is measurable with respect to $(\boldsymbol{X}_i, \boldsymbol{Z}_i)$, and Assumption 2.1 also gives $T_i \perp C_i \mid (\boldsymbol{X}_i, \boldsymbol{Z}_i)$ because ε_i is independent of $(\boldsymbol{D}_i, C_i)$. For $\mathcal{M}_1$, conditioning on the full design retains the association that can arise between the additional term $\boldsymbol{f}_i^\top \boldsymbol{\xi}_0$ and censoring after the factors are integrated out.

Define the residual-scale quantities

$$\begin{aligned} C_i^* &= \log C_i - \eta_i^0, \\ R_i &= Y_i - \eta_i^0 = \min(\varepsilon_i, C_i^*), \\ G(t \mid \boldsymbol{D}_i) &= \Pr(C_i^* \ge t \mid \boldsymbol{D}_i). \end{aligned} \tag{2.11}$$

Let $S_\varepsilon(t) = \Pr(\varepsilon_i \ge t)$.

Assumption 2.2 (A common local information interval)**.** There exist $t_\star \in \mathbb{R}$, $A > 0$, $m_\varepsilon > 0$, and $g_0 > 0$ such that, on $I = [t_\star - A, t_\star + A]$,

$$f_\varepsilon(t) \geq m_\varepsilon \quad \text{a.e.}, \qquad \inf_{t \in I} G(t \mid \boldsymbol{D}_i) \geq g_0 \quad \text{a.s.} \tag{2.12}$$

The constants used for uniform bounds are independent of n and p.

Assumption 2.2 supplies a common residual-time interval with positive event density and follow-up probability uniformly over the design, while allowing arbitrary error location. This design-uniform overlap is stronger than control of the marginal censoring proportion. For example, fixed administrative follow-up can violate the overlap condition when η_i^0 is unbounded above.

Conditioning on $D_n = \sigma(\boldsymbol{D}_1, \ldots, \boldsymbol{D}_n)$ fixes factor estimates obtained from the covariates. Patient outcomes remain conditionally independent with potentially different censoring laws. Section 4 uses this conditional structure and treats patients as the independent sampling units.

2.5 Identification of the slope target

Slope uniqueness is governed by the following sparse-design separation condition. It complements the coordinate normalization in Section 2.2.

Assumption 2.3 (No sparse affine alias)**.** For some $\kappa_{\mathrm{pop}} > 0$,

$$\mathrm{Var}(\boldsymbol{D}_i^\top \boldsymbol{v}) \geq \kappa_{\mathrm{pop}} \|\boldsymbol{v}\|_2^2 \quad \text{whenever} \quad \boldsymbol{v} = (\boldsymbol{v}_a^\top, \boldsymbol{v}_\beta^\top)^\top, \quad \|\boldsymbol{v}_\beta\|_0 \leq 2s. \tag{2.13}$$

The unrestricted block $\boldsymbol{v}_a$ allows joint variation in clinical and factor effects. Assumption 2.3 separates a sparse molecular signal from these effects; a positive lower eigenvalue of $\mathrm{Cov}(\boldsymbol{D}_i)$ is sufficient. For $\mathcal{M}_0$, the analogous restriction on $(\boldsymbol{Z}_i, \boldsymbol{X}_i)$ suffices. This condition identifies the population target, while Section 4 introduces the estimated-design cone condition for penalized estimation.

To state the target precisely without assuming integrable log-time errors, consider two independent patients, indexed by 1 and 2. Put

$$e_i^o(\boldsymbol{\vartheta}) = Y_i - \boldsymbol{D}_i^\top \boldsymbol{\vartheta}, \qquad \Theta_s = \{\boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top : \|\boldsymbol{\beta}\|_0 \leq s\}.$$

The superscript o denotes use of the latent design. Define the expected Gehan *contrast*

$$\mathcal{G}(\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0) = E\Big[\Delta_1 \Big\{ \big(e_2^o(\boldsymbol{\vartheta}) - e_1^o(\boldsymbol{\vartheta})\big)_+ - \big(e_2^o(\boldsymbol{\vartheta}_0) - e_1^o(\boldsymbol{\vartheta}_0)\big)_+ \Big\}\Big]. \tag{2.14}$$

The subtraction is inside the expectation. The 1-Lipschitz positive-part function bounds the integrand by $|(\boldsymbol{D}_1 - \boldsymbol{D}_2)^\top(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)|$, which is integrable under the covariate moment condition. Thus the Gehan contrast is well defined without an integrable log-time error (Cai et al., 2009; Suder and Molstad, 2022).

Proposition 2.1 (Separation of the rank target)**.** *Under Assumptions 2.1–2.3, let*

$$c_{\mathrm{loc}} = A m_\varepsilon^2 g_0^2, \qquad \rho_A(t) = \begin{cases} t^2, & |t| \le A, \\ 2A|t| - A^2, & |t| > A. \end{cases}$$

For every $\boldsymbol{\vartheta} \in \Theta_s$, with $\boldsymbol{v} = \boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0$,

$$\mathcal{G}(\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0) \ge \frac{c_{\mathrm{loc}}}{2} E\big[\rho_A\{(\boldsymbol{D}_1 - \boldsymbol{D}_2)^\top \boldsymbol{v}\}\big]. \tag{2.15}$$

Consequently $\boldsymbol{\vartheta}_0$ is the unique minimizer of this contrast over Θ_s for the fixed latent representation. In $\mathcal{M}_0$, the analogous contrast formed with residuals $Y_i - \boldsymbol{Z}_i^\top \boldsymbol{\theta} - \boldsymbol{X}_i^\top \boldsymbol{\beta}$ identifies $(\boldsymbol{\theta}_0, \boldsymbol{\beta}_0)$ from the observed-data law, under the corresponding observed-design separation condition.

A proof is given in Appendix A. The lower bound uses ρ_A, which is quadratic for $|t| \le A$ and linear beyond that range. Section 4 develops the corresponding finite-sample bounds.

The primary observed-data target is $(\boldsymbol{\theta}_0, \boldsymbol{\beta}_0)$ in $\mathcal{M}_0$. Extension results for $\mathcal{M}_1$ apply to a specified factor-model sequence with explicit recovery conditions; identification across unrestricted latent decompositions requires further restrictions.

2.6 Centering and error criteria

Let $\boldsymbol{X}, \boldsymbol{F}, \boldsymbol{U}, \boldsymbol{Z}$ collect row vectors $\boldsymbol{X}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top, \boldsymbol{Z}_i^\top$, respectively, and set

$$\begin{gathered} \boldsymbol{J} = \boldsymbol{I}_n - \mathbf{1}_n \mathbf{1}_n^\top / n, \\ \boldsymbol{X}_c = \boldsymbol{J}\boldsymbol{X}, \quad \boldsymbol{F}_c = \boldsymbol{J}\boldsymbol{F}, \quad \boldsymbol{U}_c = \boldsymbol{J}\boldsymbol{U}, \quad \boldsymbol{Z}_c = \boldsymbol{J}\boldsymbol{Z}. \end{gathered} \tag{2.16}$$

Then $\boldsymbol{X}_c = \boldsymbol{F}_c\boldsymbol{B}^\top + \boldsymbol{U}_c$, and

$$\begin{gathered}\boldsymbol{\eta}_c^0 := \boldsymbol{J}\boldsymbol{\eta}^0 = \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \boldsymbol{U}_c\boldsymbol{\beta}_0 + \boldsymbol{F}_c\boldsymbol{\gamma}_0, \\ Y_i - (\eta_i^0 - \bar{\eta}^0) = R_i + \bar{\eta}^0.\end{gathered} \tag{2.17}$$

The common constant $\bar{\eta}^0$ cancels from each pairwise residual difference. Sample covariate centering therefore preserves the rank comparisons and slope target while accommodating arbitrary error location.

The primary error quantities are $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2$ and $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1$. The separate quantity $n^{-1/2}\|J(\widehat{\boldsymbol{\eta}} - \boldsymbol{\eta}^0)\|_2$ measures location-invariant prediction error on the training sample. Section 3 defines the feasible criterion and reference coordinates for these targets.

3 Estimation method

The estimator uses the observed data and the centering conventions of Section 2. Write $q = d + K$, $P = p + q$, $\boldsymbol{J} = \boldsymbol{I}_n - \mathbf{1}_n\mathbf{1}_n^\top/n$, $\boldsymbol{X}_c = \boldsymbol{J}\boldsymbol{X}$, and $\boldsymbol{Z}_c = \boldsymbol{J}\boldsymbol{Z}$. Its primary target is the sparse molecular slope in $\mathcal{M}_0$, with $\mathcal{M}_1$ interpreted relative to its specified latent representation.

3.1 Covariate-only factor estimation

Take $n \geq 2$ and a prespecified $K \leq \operatorname{rank}(\boldsymbol{X}_c) \leq n - 1$. For $K > 0$, let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_K$ be orthonormal eigenvectors of $\boldsymbol{X}_c\boldsymbol{X}_c^\top/n$ associated with its largest K eigenvalues $\widehat{\ell}_1 \geq \cdots \geq \widehat{\ell}_K > 0$, and define

$$\begin{aligned}\widehat{\boldsymbol{F}} &= \sqrt{n}\,(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_K), & \widehat{\boldsymbol{\Lambda}} &= \operatorname{diag}(\widehat{\ell}_1, \ldots, \widehat{\ell}_K), \\ \widehat{\boldsymbol{B}} &= \boldsymbol{X}_c^\top\widehat{\boldsymbol{F}}/n, & \widehat{\boldsymbol{U}} &= \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top.\end{aligned} \tag{3.1}$$

Signs and tied eigenspaces follow a deterministic covariate-measurable convention. A tie across the Kth eigenvalue can alter the selected subspace; the recovery analysis assumes spectral separation at K. The construction is outcome-free and gives

$$\begin{gathered}\widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{F}}/n = \boldsymbol{I}_K, \qquad \widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{U}} = 0, \qquad \boldsymbol{J}\widehat{\boldsymbol{F}} = \widehat{\boldsymbol{F}}, \quad \boldsymbol{J}\widehat{\boldsymbol{U}} = \widehat{\boldsymbol{U}}, \\ \widehat{\boldsymbol{B}}^\top\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{\Lambda}}, \qquad \boldsymbol{X}_c\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}\end{gathered} \tag{3.2}$$

directly. This principal-component residualization follows Fan et al. (2020); sample factor–residual orthogonality allows dependence among the residual coordinates.

Let $\widehat{\boldsymbol{f}}_i$ and $\widehat{\boldsymbol{u}}_i$ denote the column vectors corresponding to row i of $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{U}}$. Set

$$\begin{aligned} \widehat{\boldsymbol{A}} &= (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}}), \\ \widehat{\boldsymbol{W}} &= (\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{U}}), \\ \boldsymbol{a} = (\boldsymbol{\theta}^\top, \boldsymbol{\gamma}^\top)^\top, &\quad \boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top. \end{aligned} \tag{3.3}$$

Thus $\widehat{\boldsymbol{W}}$ is $n \times P$, with row $\widehat{\boldsymbol{W}}_i^\top$. The convention $K = 0$ omits the factor block and sets $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c$, defining the unadjusted comparison method.

Residual coefficients retain the molecular coordinate scales fixed in Section 2. Under a rescaling $\widetilde{\boldsymbol{U}} = \widehat{\boldsymbol{U}}\boldsymbol{S}^{-1}$ with positive diagonal $\boldsymbol{S} = \mathrm{diag}(s_1, \ldots, s_p)$, the equivalent coordinates and penalty are $\boldsymbol{\zeta} = \boldsymbol{S}\boldsymbol{\beta}$ and $\lambda \sum_j |\zeta_j|/s_j$. All learned transformations are confined to the training sample.

3.2 Censored pairwise loss and scalar smoothing

For a finite parameter vector, define

$$\begin{aligned} e_i(\boldsymbol{\vartheta}) &= Y_i - \widehat{\boldsymbol{W}}_i^\top \boldsymbol{\vartheta}, \\ r_{ij}(\boldsymbol{\vartheta}) &= e_j(\boldsymbol{\vartheta}) - e_i(\boldsymbol{\vartheta}), \\ \boldsymbol{w}_{ij} &= \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j, \\ N &= n(n-1). \end{aligned} \tag{3.4}$$

The unsmoothed Gehan loss is

$$L_{n,0}(\boldsymbol{\vartheta}) = \frac{1}{N} \sum_{i \neq j} \Delta_i \{ r_{ij}(\boldsymbol{\vartheta}) \}_+. \tag{3.5}$$

The criterion follows regularized Gehan estimation (Cai et al., 2009; Suder and Molstad, 2022). Event patients anchor the index i, and every other patient, including censored patients, contributes as j. We retain the ordered off-diagonal normalization $N = n(n-1)$ throughout.

Let $V \sim N(0,1)$ be an auxiliary integration variable, unrelated to the AFT error. For a common scalar bandwidth $h > 0$, define

$$\begin{aligned} \ell_h(r) &= E_V(r + hV)_+ = r\Phi(r/h) + h\phi(r/h), \\ L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N}\sum_{i \neq j} \Delta_i \ell_h\{r_{ij}(\boldsymbol{\vartheta})\}, \end{aligned} \tag{3.6}$$

where ϕ and Φ are the standard normal density and distribution function. The Gaussian convolution is evaluated analytically while the AFT error distribution remains unspecified. Differentiation gives

$$\ell_h'(r) = \Phi(r/h), \qquad \ell_h''(r) = h^{-1}\phi(r/h) > 0. \tag{3.7}$$

Consequently ℓ_h is convex and 1-Lipschitz, with derivative $1/2$ at zero.

Proposition 3.1 (Uniform loss approximation)**.** *For every $r \in \mathbb{R}$ and $h > 0$,*

$$\begin{aligned} 0 \leq \ell_h(r) - r_+ &\leq \frac{h}{\sqrt{2\pi}}, \\ \int_{\mathbb{R}} |\Phi(r/h) - \mathbf{1}\{r \geq 0\}| \, \mathrm{d}r &= h\sqrt{\frac{2}{\pi}}. \end{aligned} \tag{3.8}$$

For any fixed observed sample, writing $\bar{\Delta} = n^{-1}\sum_i \Delta_i$,

$$0 \leq L_{n,h}(\boldsymbol{\vartheta}) - L_{n,0}(\boldsymbol{\vartheta}) \leq \frac{\bar{\Delta} h}{\sqrt{2\pi}} \quad \textit{uniformly over } \boldsymbol{\vartheta} \in \mathbb{R}^P. \tag{3.9}$$

Appendix B proves the deterministic identities in this section. Proposition 3.1 controls the loss approximation; Section 4 combines bandwidth-dependent score bias, identification, and curvature to obtain coefficient bounds.

Scalar smoothing of censored rank scores has precedents in Heller (2007). It uses a common bandwidth, whereas the parameter-perturbation construction of Johnson and Strawderman (2009) induces pair-specific bandwidths. For a perturbation $n^{-1/2}\boldsymbol{\Gamma}\boldsymbol{V}_P$, with $\boldsymbol{V}_P \sim N(0, \boldsymbol{I}_P)$, these are

$$h_{ij} = n^{-1/2}\{\boldsymbol{w}_{ij}^\top \boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top \boldsymbol{w}_{ij}\}^{1/2},$$

which generally varies by pair. The common scalar h used here leads to the smoothing-bias analysis in Section 4.

3.3 Partial regularization and finite minimizers

For fixed $h > 0$ and $\lambda > 0$, let

$$\begin{aligned} \mathcal{S}_{n,h,\lambda} &= \arg\min_{\vartheta \in \mathbb{R}^P} Q_{n,h,\lambda}(\boldsymbol{\vartheta}), \\ Q_{n,h,\lambda}(\boldsymbol{\vartheta}) &= L_{n,h}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\beta}\|_1. \end{aligned} \tag{3.10}$$

When this set is nonempty, $\widehat{\boldsymbol{\vartheta}}$ denotes a member. The ℓ_1 penalty produces sparsity in the p molecular coefficients, and clinical and factor effects are estimated jointly with them. Section 4 establishes coefficient-error guarantees.

The identity $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ implies

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{\vartheta} &= \boldsymbol{Z}_c\boldsymbol{\theta} + \boldsymbol{X}_c\boldsymbol{\beta} + \widehat{\boldsymbol{F}}\boldsymbol{\nu}, \\ \boldsymbol{\nu} &= \boldsymbol{\gamma} - \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}. \end{aligned} \tag{3.11}$$

For fixed training data, (3.11) is a bijective reparameterization preserving $\|\boldsymbol{\beta}\|_1$. The observed-$\boldsymbol{X}$ formulation includes a freely estimated principal-component term. Setting $\boldsymbol{\gamma} = \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}$ recovers the ordinary penalized smoothed AFT fit; allowing $\boldsymbol{\gamma}$ to vary gives the proposed relaxation. Both blocks are optimized jointly under the nonlinear pairwise loss.

Finite attainment with the unpenalized block is governed by the nuisance-design geometry. A sufficient sample condition, when $q > 0$, is

$$\chi_n := \inf_{\|\boldsymbol{v}\|_2 = 1} \frac{1}{N} \sum_{i \neq j} \Delta_i \{(\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j)^\top \boldsymbol{v}\}_+ > 0. \tag{3.12}$$

Condition (3.12) makes $\mathcal{S}_{n,h,\lambda}$ nonempty and compact; Appendix B.3 proves this result and illustrates nonattainment. For $q = 0$, the positive ℓ_1 penalty bounds the sublevel sets. Section 4 also establishes finite attainment on the statistical event used for estimation bounds.

A deterministic tie-breaking rule may select among multiple minimizers; numerical eligibility is recorded separately.

3.4 Reference coordinates and the approximation term

To relate (3.10) to the target, write the centered latent matrices as $\boldsymbol{F}_c = \boldsymbol{J}\boldsymbol{F}$ and $\boldsymbol{U}_c = \boldsymbol{J}\boldsymbol{U}$. For any invertible alignment matrix $\boldsymbol{H}$, set

$$\begin{aligned} \boldsymbol{E}_F &= \widehat{\boldsymbol{F}} - \boldsymbol{F}_c\boldsymbol{H}^\top, \\ \boldsymbol{\gamma}^\dagger &= \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}_0 + \boldsymbol{H}^{-T}\boldsymbol{\xi}_0, \\ \boldsymbol{\vartheta}^\dagger &= (\boldsymbol{\theta}_0^\top, (\boldsymbol{\gamma}^\dagger)^\top, \boldsymbol{\beta}_0^\top)^\top. \end{aligned} \tag{3.13}$$

These reference coordinates link the feasible criterion to the population target. In $\mathcal{M}_0$, $\boldsymbol{\gamma}^\dagger = \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}_0$ and the alignment matrix drops out. Recall

$$\boldsymbol{\eta}^0 = \boldsymbol{Z}\boldsymbol{\theta}_0 + \boldsymbol{X}\boldsymbol{\beta}_0 + \boldsymbol{F}\boldsymbol{\xi}_0, \qquad \boldsymbol{\eta}_c^0 = \boldsymbol{J}\boldsymbol{\eta}^0, \qquad R_i = Y_i - \eta_i^0.$$

Proposition 3.2 (Exact reference representation)**.** *With* $\mathbf{d} = \boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{\xi}_0$ *and* $b_n = \|\mathbf{d}\|_\infty$*,*

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{\vartheta}^\dagger &= \boldsymbol{\eta}_c^0 + \mathbf{d}, & \boldsymbol{J}\mathbf{d} &= \mathbf{d}, \\ e_i(\boldsymbol{\vartheta}^\dagger) &= R_i + \bar{\eta}^0 - d_i, & r_{ij}(\boldsymbol{\vartheta}^\dagger) &= R_j - R_i + d_i - d_j. \end{aligned} \tag{3.14}$$

Moreover,

$$\begin{aligned} b_n &\le \left(\max_i \|(\boldsymbol{E}_F)_{i\cdot}\|_2\right) \|\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\|_2, \\ \widehat{\boldsymbol{W}}\hat{\boldsymbol{\vartheta}} - \boldsymbol{\eta}_c^0 &= \widehat{\boldsymbol{W}}(\hat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger) + \mathbf{d}. \end{aligned} \tag{3.15}$$

In $\mathcal{M}_0$*,* $\mathbf{d} = 0$ *exactly for every admissible PCA output.*

The common location in (3.14) cancels within each pair. Thus b_n measures approximation of the additional factor contribution. For $\mathcal{M}_0$, the ordinary predictor is represented exactly and factor estimation enters through the estimated-design geometry. Nuisance error is measured from the sample-dependent reference $\boldsymbol{\gamma}^\dagger$, allowing $\boldsymbol{B}^\top\boldsymbol{\beta}_0$ to grow with the sparse signal.

Within a fixed estimated subspace, replacing $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{B}}$ by $\widehat{\boldsymbol{F}}\boldsymbol{Q}^\top$ and $\widehat{\boldsymbol{B}}\boldsymbol{Q}^\top$, and $\boldsymbol{\gamma}$ by $\boldsymbol{Q}\boldsymbol{\gamma}$, preserves $\widehat{\boldsymbol{U}}$, the predictor, and the objective. The fitted-predictor and molecular-minimizer sets are invariant under this coordinate rotation. Subspace selection at a spectral tie is governed by the convention in Section 3.1.

3.5 Training-frozen scores for held-out patients

For a new patient, the training means and loadings define a frozen prediction map. Let $\boldsymbol{x}_c = \boldsymbol{x} - \bar{\boldsymbol{X}}$ and $\boldsymbol{z}_c = \boldsymbol{z} - \bar{\boldsymbol{Z}}$. For $K > 0$, set

$$\begin{aligned} \widehat{\boldsymbol{f}}(\boldsymbol{x}) &= \widehat{\boldsymbol{\Lambda}}^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_c, \\ \widehat{\boldsymbol{u}}(\boldsymbol{x}) &= \boldsymbol{x}_c - \widehat{\boldsymbol{B}}\widehat{\boldsymbol{f}}(\boldsymbol{x}), \\ \widehat{\eta}(\boldsymbol{x},\boldsymbol{z}) &= \boldsymbol{z}_c^{\top}\widehat{\boldsymbol{\theta}} + \widehat{\boldsymbol{f}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\gamma}} + \widehat{\boldsymbol{u}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\beta}}. \end{aligned} \tag{3.16}$$

For $K = 0$, omit the factor terms. By (3.2), this least-squares projection onto the estimated loading space reproduces the fitted training factors and residuals exactly.

Equivalently,

$$\begin{aligned} \widehat{\eta}(\boldsymbol{x},\boldsymbol{z}) &= \boldsymbol{z}_c^{\top}\widehat{\boldsymbol{\theta}} + \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{b}}_{\mathrm{pred}}, \\ \widehat{\boldsymbol{b}}_{\mathrm{pred}} &= \widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}(\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{\beta}}). \end{aligned} \tag{3.17}$$

The sparse coefficient $\widehat{\boldsymbol{\beta}}$ summarizes residual effects, whereas $\widehat{\boldsymbol{b}}_{\mathrm{pred}}$ combines residual and factor-mediated contributions and can be dense. Higher $\widehat{\eta}$ denotes longer conditional event times; $-\widehat{\eta}$ supplies the opposite risk ordering. The score represents relative log-time location. Calibrated event-time quantiles additionally require the error distribution and its location.

3.6 Bandwidth, penalty, and training selection

The estimator is indexed by (K, h, λ). The coefficient analysis takes the correct K as fixed, with $h = h_n > 0$ and $\lambda = \lambda_n > 0$ deterministic or fixed after conditioning on the design. Section 4 relates the smoothing bias, b_n, and stochastic fluctuations to the penalty and gives sufficient bandwidth and penalty conditions for the stated error rates.

For empirical tuning, let $\mathcal{V}_1, \ldots, \mathcal{V}_L$ be held-out folds of sizes $m_\ell \geq 2$. For every candidate (K, h, λ), all learned preprocessing, PCA, and coefficient fitting use the complement of $\mathcal{V}_\ell$. Apply (3.16)

to obtain $\widehat{\eta}_i^{(-\ell)}$ and put $e_i^{(-\ell)} = Y_i - \widehat{\eta}_i^{(-\ell)}$. The cross-validation (CV) score is

$$\mathrm{CV}(K,h,\lambda) = \frac{\displaystyle\sum_{\ell=1}^{L} \sum_{\substack{i,j\in\mathcal{V}_\ell \\ i\neq j}} \Delta_i \left[\{e_j^{(-\ell)} - e_i^{(-\ell)}\}_+ - \{Y_j - Y_i\}_+\right]}{\displaystyle\sum_{\ell=1}^{L} m_\ell(m_\ell - 1)}. \tag{3.18}$$

The candidate-independent baseline subtraction preserves the selection while expressing validation as a finite loss contrast. The same unsmoothed score is used for every candidate bandwidth.

Within-fold differences remove arbitrary fold-specific locations. All candidate bandwidths share the same unsmoothed score, computed over within-fold pairs. Alternative pooling rules are studied by Suder and Molstad (2022). Section 6.4 specifies numerical eligibility and tie handling for the empirical study. The coefficient theorems use fixed or covariate-measurable tuning; the outcome-selected procedure is evaluated separately.

3.7 Derivative and optimality interface

For $h > 0$, differentiation of the finite sum gives

$$\begin{aligned} \nabla L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i\neq j} \Delta_i \Phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \boldsymbol{w}_{ij}, \\ \nabla^2 L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{Nh} \sum_{i\neq j} \Delta_i \phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top. \end{aligned} \tag{3.19}$$

The positive-semidefinite Hessian establishes convexity. A finite vector is a global minimizer of (3.10) if and only if

$$\begin{aligned} \nabla_a L_{n,h}(\widehat{\boldsymbol{\vartheta}}) &= 0, \\ 0 &\in \nabla_\beta L_{n,h}(\widehat{\boldsymbol{\vartheta}}) + \lambda \partial \|\widehat{\boldsymbol{\beta}}\|_1. \end{aligned} \tag{3.20}$$

A nonzero molecular coordinate has gradient $-\lambda \operatorname{sign}(\widehat{\beta}_j)$, and a zero coordinate has gradient in $[-\lambda, \lambda]$. Numerical stationarity is assessed on these coordinates and the complete unpenalized block.

The gradient admits an efficient matrix form. Define $V_{ii} = 0$, $V_{ij} = \Delta_i \Phi(r_{ij}/h)$ for $i \neq j$, and $c_i = \sum_j V_{ij} - \sum_j V_{ji}$. Then

$$\nabla L_{n,h}(\boldsymbol{\vartheta}) = \widehat{\boldsymbol{W}}^\top \boldsymbol{c}/N. \tag{3.21}$$

4 Theoretical properties

The analysis combines conditional score control, local event information, and estimated-design geometry to establish finite-sample coefficient bounds. For the ordinary model $\mathcal{M}_0$, the reference predictor is exact. Primitive factor conditions then supply the design event, and the extension to $\mathcal{M}_1$ adds a separate approximation term. Pairing and contraction (Clémençon et al., 2008) and localized regularization (Negahban et al., 2012) provide the analytic tools.

4.1 Conditional setup and design geometry

Write $q = d + K$, $P = p + q$, $k = s + q \geq 1$, $N = n(n-1)$, and condition on $D_n = \sigma(\boldsymbol{D}_1, \ldots, \boldsymbol{D}_n)$, where $\boldsymbol{D}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top)^\top$. The factor estimates, centering, and tuning in this analysis are covariate-measurable. Outcomes are conditionally independent with patient-specific conditional laws. The criterion remains

$$\begin{aligned} Q(\boldsymbol{\vartheta}) &= L_{n,h}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\beta}\|_1, \\ L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i \neq j} \Delta_i \ell_h\{Y_j - Y_i + \boldsymbol{w}_{ij}^\top \boldsymbol{\vartheta}\}, \end{aligned} \tag{4.1}$$

where $\ell_h(t) = t\Phi(t/h) + h\phi(t/h)$, $\widehat{\boldsymbol{W}} = (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}}, \widehat{\boldsymbol{U}})$ is centered, and $\boldsymbol{w}_{ij} = \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j$. In $\mathcal{M}_0$, the reference point is

$$\begin{aligned} \boldsymbol{\vartheta}^\dagger &= (\boldsymbol{\theta}_0^\top, (\widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0)^\top, \boldsymbol{\beta}_0^\top)^\top, \\ \widehat{\boldsymbol{W}} \boldsymbol{\vartheta}^\dagger &= \boldsymbol{\eta}_c^0. \end{aligned} \tag{4.2}$$

Conditioning fixes the sample-dependent reference. Its representation accommodates unrestricted $\|\boldsymbol{B}^\top \boldsymbol{\beta}_0\|_2$.

Two sample scales are kept distinct:

$$V_n = \max_{j \le P} \frac{\|\widehat{\boldsymbol{W}}_{\cdot j}\|_2}{\sqrt{n}}, \qquad M_n = \max_{i,j} |\widehat{W}_{ij}|, \qquad \widehat{\boldsymbol{\Gamma}} = \frac{1}{N} \sum_{i \neq j} \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top = \frac{2}{n-1} \widehat{\boldsymbol{W}}^\top \widehat{\boldsymbol{W}}. \tag{4.3}$$

The column scale V_n controls stochastic fluctuations and score bias. The larger entry envelope M_n is used only to keep predictors inside a common local information interval. For $\boldsymbol{v} = (\boldsymbol{v}_a^\top, \boldsymbol{v}_\beta^\top)^\top$, define

$$\mathcal{C}(S) = \{\boldsymbol{v} : \|(\boldsymbol{v}_\beta)_{S^c}\|_1 \le 3(\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1)\}. \tag{4.4}$$

On this cone, $\|\boldsymbol{v}\|_1 \le 4\sqrt{k}\|\boldsymbol{v}\|_2$. Thus the unpenalized block contributes to the effective dimension.

Assumption 4.1 (Conditional restricted design)**.** The covariate realization satisfies, for some $\kappa_X > 0$,

$$\boldsymbol{v}^\top \widehat{\boldsymbol{\Gamma}} \boldsymbol{v} \ge \kappa_X \|\boldsymbol{v}\|_2^2 \quad \text{for every } \boldsymbol{v} \in \mathcal{C}(S). \tag{4.5}$$

Assumption 4.1 controls the estimated design on the error cone $\mathcal{C}(S)$, the geometry relevant to penalized estimation.

Assumptions 2.1–2.2 specify the common error law, conditional censoring, bounded error density, and a design-uniform local information interval. They permit censoring atoms and log-time errors without moments. Put

$$c_{\mathrm{loc}} = A m_\varepsilon^2 g_0^2, \qquad c_\phi = 2\Phi(1) - 1, \qquad \kappa = c_{\mathrm{loc}} c_\phi \kappa_X. \tag{4.6}$$

4.2 Score bias, pairwise fluctuations, and curvature

Let $R_i = Y_i - \eta_i^0$. To cover both models in a single set of analytic lemmas, allow a covariate-measurable centered vector $\mathbf{d}$ with $\|\mathbf{d}\|_\infty \le b$, such that

$$\widehat{\boldsymbol{W}} \boldsymbol{\vartheta}^\dagger = \boldsymbol{\eta}_c^0 + \mathbf{d}, \qquad r_{ij}(\boldsymbol{\vartheta}^\dagger) = R_j - R_i + d_i - d_j.$$

For $\mathcal{M}_0$, take $b = 0$. Define the conditional risk *contrast* and mean score by

$$\begin{aligned} \mathcal{R}_h(\boldsymbol{v}) &= E\{L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger) \mid D_n\}, \\ \boldsymbol{g}_h &= E\{\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger) \mid D_n\}. \end{aligned} \tag{4.7}$$

The loss increment is bounded in absolute value by $2V_n\|\boldsymbol{v}\|_1$, making (4.7) integrable without an error-moment assumption.

Lemma 4.1 (Score and contrast control)**.** *Suppose the common-error and conditional-censoring assumptions hold. Fix* $0 < \alpha < 1/4$ *and* $r > 0$ *after conditioning, and set*

$$t_\alpha = \sqrt{\frac{\log(2P/\alpha)}{n}}, \qquad \mu(b,h) = 2V_n L_\varepsilon\{2b + h\sqrt{2/\pi}\}. \tag{4.8}$$

Then $\|\boldsymbol{g}_h\|_\infty \le \mu(b,h)$*. Each of the following inequalities holds with conditional probability at least* $1-\alpha$*:*

$$\|\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger) - \boldsymbol{g}_h\|_\infty \le \sqrt{8}\, V_n t_\alpha, \tag{4.9}$$

$$\sup_{\|\boldsymbol{v}\|_1 \le r} |L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger) - \mathcal{R}_h(\boldsymbol{v})| \le 96 V_n r t_\alpha. \tag{4.10}$$

The two events hold jointly with probability at least $1-2\alpha$*.*

The stochastic scale is $V_n\sqrt{\log(2P/\alpha)/n}$: patients are the independent sampling units, and the bound is uniform in h. Averaging over all pairings retains patient-specific censoring laws. Rademacher moment contraction (Clémençon et al., 2008; Ledoux and Talagrand, 1991) yields column-scale control in Appendix C.3.

Lemma 4.2 (Conditional local curvature)**.** *Under the outcome conditions and Assumption 4.1, if* $h \le A/2$ *and*

$$2b + 2M_n\|\boldsymbol{v}\|_1 \le A/2, \tag{4.11}$$

then, for $\boldsymbol{v} \in \mathcal{C}(S)$*,*

$$\mathcal{R}_h(\boldsymbol{v}) - \boldsymbol{g}_h^\top \boldsymbol{v} \ge \frac{c_{\mathrm{loc}} c_\phi}{2} \boldsymbol{v}^\top \widehat{\boldsymbol{\Gamma}} \boldsymbol{v} \ge \frac{\kappa}{2}\|\boldsymbol{v}\|_2^2. \tag{4.12}$$

Lemma 4.2 supplies curvature of the conditional expected contrast on $\mathcal{C}(S)$, while Lemma 4.1 controls its empirical fluctuation. The event-weighted difference density permits censoring-induced atoms.

4.3 Ordinary-model estimation and existence

Theorem 4.1 (Conditional guarantee in $\mathcal{M}_0$)**.** *Suppose $n \geq 2$, the ordinary model and outcome conditions hold, and the conditioned design satisfies Assumption 4.1. For $0 < \alpha < 1/4$, choose $h > 0$, $\lambda > 0$, and*

$$\lambda \geq 4\mu(0, h) + 384 V_n t_\alpha, \qquad R = \frac{8\sqrt{k}\lambda}{\kappa}. \tag{4.13}$$

If

$$h \leq A/2, \qquad 8M_n\sqrt{k}R \leq A/2, \tag{4.14}$$

then, with conditional probability at least $1 - 2\alpha$, the set of global minimizers of Q is nonempty and compact. Every such minimizer satisfies

$$\begin{aligned} \|\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 &\leq \frac{8\sqrt{k}\lambda}{\kappa}, \\ \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 &\leq \frac{32k\lambda}{\kappa}. \end{aligned} \tag{4.15}$$

In particular the first bound also controls $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2$.

Theorem 4.1 establishes finite attainment and coefficient bounds on the same statistical event. Appendix C.5 localizes the sublevel set through $\boldsymbol{\vartheta}^\dagger$ inside a compact ball.

The localized basic inequality follows the decomposable-regularization framework (Negahban et al., 2012) with a seminorm penalty on the full parameter. Including the nuisance directions yields the sharper cone relation

$$\|(\boldsymbol{v}_\beta)_{S^c}\|_1 \leq 3\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1. \tag{4.16}$$

The nuisance error is $\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0$, measured in the estimated-factor coordinates. The sparse molecular coefficient and the deployment slope retain their distinct definitions from Section 3.

4.4 From primitive factor conditions to estimated-design geometry

The primitive conditions concern the joint covariate distribution. They supply the estimated-design geometry separately from the assumptions on ε_i.

Assumption 4.2 (Independent jointly sub-Gaussian factor design)**.** For fixed $K \geq 1$ and $d \geq 0$, the centered $\boldsymbol{D}_i$ are independent and identically distributed and satisfy

$$\begin{aligned} c_D \boldsymbol{I}_P \preceq \boldsymbol{\Sigma}_D := E(\boldsymbol{D}_i \boldsymbol{D}_i^\top) \preceq C_D \boldsymbol{I}_P, \\ \sup_{\|\boldsymbol{v}\|_2 = 1} \|\boldsymbol{v}^\top \boldsymbol{D}_i\|_{\psi_2} \leq L_D. \end{aligned} \tag{4.17}$$

Here $\|U\|_{\psi_2} = \inf\{t > 0 : E \exp(U^2/t^2) \leq 2\}$. Also $E(\boldsymbol{f}_i \boldsymbol{f}_i^\top) = \boldsymbol{I}_K$, $E(\boldsymbol{f}_i \boldsymbol{u}_i^\top) = 0$, and the deterministic loading matrix obeys

$$c_B \boldsymbol{I}_K \preceq \boldsymbol{B}^\top \boldsymbol{B}/p \preceq C_B \boldsymbol{I}_K, \qquad \max_{j \leq p} \|\boldsymbol{B}_{j\cdot}\|_2 \leq L_B. \tag{4.18}$$

All displayed constants are uniform in n, p.

The covariance lower bound and factor pervasiveness jointly specify a sufficient design regime. The normalization and alignment follow the PCA framework of Bai and Ng (2013) and Fan et al. (2020); Assumption 4.2 supports the norm bounds below.

For $0 < \delta < 1/4$, put

$$\begin{aligned} &L_\delta = \log(64nP/\delta), \quad t_\delta = \sqrt{L_\delta/n}, \quad \tau_\delta = t_\delta + L_\delta/n, \\ &a_\delta = \tau_\delta + p^{-1/2}, \quad r_{F,\delta} = n^{-1/2} + \sqrt{L_\delta/p}. \end{aligned} \tag{4.19}$$

The quantity χ_n in Section 3 continues to denote the separate sample-existence criterion.

Proposition 4.1 (PCA and design event)**.** *Under Assumption 4.2, there are fixed positive constants c, C such that, if $n \wedge p \geq C L_\delta$, an event $\mathcal{E}_D$ has probability at least $1 - \delta$ and the following properties. The selected spectrum is separated at K, $cp\boldsymbol{I}_K \preceq \widehat{\boldsymbol{\Lambda}} \preceq Cp\boldsymbol{I}_K$, and*

$$\begin{aligned} \boldsymbol{H} = \widehat{\boldsymbol{\Lambda}}^{-1} \widehat{\boldsymbol{B}}^\top \boldsymbol{B}, \\ \boldsymbol{E}_F = \widehat{\boldsymbol{F}} - \boldsymbol{F}_c \boldsymbol{H}^\top = \boldsymbol{U}_c \widehat{\boldsymbol{B}} \widehat{\boldsymbol{\Lambda}}^{-1}, \\ \|\boldsymbol{H}\|_{\mathrm{op}} + \|\boldsymbol{H}^{-1}\|_{\mathrm{op}} \leq C. \end{aligned} \tag{4.20}$$

On this event,

$$\begin{aligned} \|\boldsymbol{E}_F\|_{\mathrm{op}}/\sqrt{n} \leq C(n^{-1/2} + p^{-1/2}), \\ \max_i \|(\boldsymbol{E}_F)_{i\cdot}\|_2 \leq C r_{F,\delta}. \end{aligned} \tag{4.21}$$

$$\begin{aligned} \max_j \|\widehat{\boldsymbol{U}}_{\cdot j} - \boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n} \le C a_\delta, \\ V_n \le C, \qquad M_n \le C\sqrt{L_\delta}. \end{aligned} \tag{4.22}$$

If, additionally,

$$k\tau_\delta \le c, \qquad \sqrt{k}a_\delta \le c, \tag{4.23}$$

then Assumption 4.1 holds with a fixed lower bound $\kappa_X \ge c_X > 0$. If Assumption 4.2 is supplemented by $\max_j |D_{ij}| \le L_0$ almost surely for a fixed L_0, then $M_n \le C$ on the same type of event.

The operator, rowwise, and columnwise bounds support factor alignment, predictor approximation, and design transfer, respectively. Appendix D derives them through covariance concentration and sub-Gaussian net bounds (Vershynin, 2012), retaining the same-sample dependence of $\widehat{\boldsymbol{F}}$ and $\boldsymbol{U}_c$.

For clarity, the deterministic transfer step is stated separately. Let $\boldsymbol{W} = (\boldsymbol{Z}_c, \boldsymbol{F}_c\boldsymbol{H}^\top, \boldsymbol{U}_c)$ and define $\|\boldsymbol{x}\|_{\text{pair}}^2 = N^{-1}\sum_{i\neq j}(x_i - x_j)^2$.

Lemma 4.3 (Column-error transfer)**.** *If $\max_j \|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2/\sqrt{n} \le a$ and $\|\boldsymbol{W}\boldsymbol{v}\|_{\text{pair}} \ge \sqrt{\kappa_0}\|\boldsymbol{v}\|_2$ for $\boldsymbol{v} \in \mathcal{C}(S)$, then*

$$\begin{aligned} \|\widehat{\boldsymbol{W}}\boldsymbol{v}\|_{\text{pair}} \ge (\sqrt{\kappa_0} - 8a\sqrt{k})\|\boldsymbol{v}\|_2, \\ \boldsymbol{v} \in \mathcal{C}(S). \end{aligned} \tag{4.24}$$

Thus $8a\sqrt{k} \le \sqrt{\kappa_0}/2$ yields a positive estimated-design lower bound. Column mean-square control suffices for this restricted-design transfer.

4.5 Unconditional ordinary-model rates

Theorem 4.2 (A primitive-condition guarantee for $\mathcal{M}_0$)**.** *Suppose the ordinary model, the outcome conditions, and Assumption 4.2 hold with uniform constants. There are constants $C_*, c_* > 0$, depending only on these constants and fixed K, d, for which the following statement holds. Take $0 < \delta < 1/4$, $n \wedge p \ge C_* L_\delta$, and impose (4.23) with sufficiently small constants. Choose*

$$0 < h \le A/2, \qquad \lambda = C_*\{t_\delta + h\}, \qquad k\sqrt{L_\delta}\,\lambda \le c_*. \tag{4.25}$$

Then, with probability at least $1-3\delta$*, the minimizer set is nonempty and compact, and every minimizer satisfies*

$$\begin{aligned} \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2 &\le C\sqrt{k}(t_\delta + h), \\ \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 &\le Ck(t_\delta + h). \end{aligned} \tag{4.26}$$

The corresponding reference-coordinate full-vector bound also holds. Under the additional bounded-coordinate condition of Proposition 4.1, the final condition in (4.25) *can instead be* $k\lambda \le c_*$.

With C_* sufficiently large and c_* sufficiently small, the tuning conditions combine smoothing bias and sampling fluctuations. The probability accounts for the design event and both conditional outcome events. Taking $h \le Ct_\delta$ gives the ℓ_2 rate $\sqrt{kL_\delta/n}$ in both covariate classes. For $p \ge n$, sufficient growth conditions are

$$\begin{aligned} &k^2 L_\delta^2/n \longrightarrow 0, \text{ jointly sub-Gaussian design,} \\ &k^2 L_\delta/n \longrightarrow 0, \text{ jointly sub-Gaussian and bounded coordinates.} \end{aligned} \tag{4.27}$$

For a probability tending to one, take $\delta = \delta_n \downarrow 0$ in these sufficient growth conditions. Using the column scale V_n in the stochastic bound removes an additional logarithmic loss from the error rate; the entry envelope M_n remains in the locality requirement.

A concrete construction demonstrates compatibility with Cauchy log-time errors. Take fixed $K = 1$, independent Rademacher coordinates in $(\boldsymbol{Z}_i, \boldsymbol{f}_i, \boldsymbol{u}_i)$, and $\boldsymbol{B}_j = b_0 \ne 0$. Let the error be standard Cauchy and independent of the design, and set $\log C_i = \eta_i^0 + 2$. On $[-1, 1]$, the error density is positive and $G = 1$, while censoring has positive probability. With $p = n^2$ and $s = \lfloor n^{1/4} \rfloor$, the bounded-design condition holds for polynomially decreasing δ_n.

4.6 Additional factor effects

In $\mathcal{M}_1$, fix the latent representation described in Section 2. For a covariate-measurable invertible $\boldsymbol{H}$, let

$$\begin{aligned} \boldsymbol{\gamma}^\dagger &= \widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0 + \boldsymbol{H}^{-T} \boldsymbol{\xi}_0, \\ \mathbf{d} &= \boldsymbol{E}_F \boldsymbol{H}^{-T} \boldsymbol{\xi}_0, \\ b_n &= \|\mathbf{d}\|_\infty. \end{aligned} \tag{4.28}$$

These reference quantities isolate approximation of the additional factor contribution.

Corollary 4.1 (Conditional and primitive-design extensions)**.** *Theorem 4.1 holds for this reference point after replacing* $\mu(0, h)$ *by* $\mu(b_n, h)$ *and replacing* (4.14) *by*

$$h \le A/2, \qquad 2b_n + 8M_n\sqrt{k}R \le A/2. \tag{4.29}$$

Under the primitive design assumptions and $\|\boldsymbol{\xi}_0\|_2 \le \Xi$*, Proposition 4.1 yields* $b_n \le C\Xi r_{F,\delta}$*. In the unconditional statement take*

$$\lambda = C_*\{t_\delta + h + \Xi r_{F,\delta}\}, \qquad \Xi r_{F,\delta} + k\sqrt{L_\delta}\lambda \le c_*. \tag{4.30}$$

The ℓ_2 *and* ℓ_1 *bounds are respectively*

$$C\sqrt{k}\{t_\delta + h + \Xi r_{F,\delta}\}, \qquad Ck\{t_\delta + h + \Xi r_{F,\delta}\}, \tag{4.31}$$

with probability at least $1 - 3\delta$*. With bounded coordinates replace* $k\sqrt{L_\delta}\lambda$ *by* $k\lambda$*. Other primitive-design and bandwidth conditions remain in force.*

Setting $\boldsymbol{\xi}_0 = 0$ deletes the approximation term exactly. For $p \ge n$ and bounded Ξ, $r_{F,\delta} \lesssim t_\delta$. These extension bounds retain the specified latent representation.

The coefficient guarantees apply to fixed, correct K and covariate-measurable tuning under the stated error, censoring, and design conditions. Section 5 connects them to finite numerical accuracy; Section 6 studies the separate outcome-based selection procedure.

5 Computation

We solve the partially penalized smoothed-rank objective in the original coefficient coordinates. For a fixed training sample and

(K, h, λ) with $h, \lambda > 0$, write $\widehat{\boldsymbol{W}} = (\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{U}})$, $\widehat{\boldsymbol{A}} = (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}})$, $q = d+K$, $P = q + p$, and $\boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top$. The objective is

$$\begin{aligned} Q(\boldsymbol{\vartheta}) &= f_h(\boldsymbol{\vartheta}) + g_\lambda(\boldsymbol{\vartheta}), \\ g_\lambda(\boldsymbol{\vartheta}) &= \lambda \|\boldsymbol{\beta}\|_1, \\ f_h(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i \neq j} \Delta_i \ell_h(r_{ij}), \\ r_{ij} &= Y_j - Y_i + (\widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j)^\top \boldsymbol{\vartheta}, \\ N &= n(n-1). \end{aligned} \tag{5.1}$$

where Y_i is already on the log-time scale and $\ell_h(r) = r\Phi(r/h) + h\phi(r/h)$. All arrays are finite, $n \geq 2$, and $\Delta_i \in \{0, 1\}$. PCA and preprocessing remain fixed during a solve.

The reference algorithm is monotone proximal gradient with backtracking. The empirical implementation combines acceleration and qualified second-order refinement. These algorithms specialize composite-convex optimization (Beck and Teboulle, 2009; Parikh and Boyd, 2014) to the pairwise loss and unpenalized block in (5.1). Scalable alternatives for the unsmoothed AFT criterion include the prox-linear alternating direction method of multipliers (ADMM) algorithm of Suder and Molstad (2022).

5.1 Blocked pairwise evaluation

Compute $\boldsymbol{e} = \boldsymbol{Y} - \widehat{\boldsymbol{W}}\boldsymbol{\vartheta}$. For $i \neq j$, put $V_{ij} = \Delta_i \Phi\{(e_j - e_i)/h\}$, and set $V_{ii} = 0$. The gradient is

$$c_i = \sum_j V_{ij} - \sum_j V_{ji}, \qquad \nabla f_h(\boldsymbol{\vartheta}) = \widehat{\boldsymbol{W}}^\top \boldsymbol{c}/N. \tag{5.2}$$

The coefficients satisfy $\mathbf{1}_n^\top \boldsymbol{c} = 0$. Event patients anchor rows i, all other patients contribute as j, and the diagonal is excluded from the ordered-pair sum.

For an event-row block $\mathcal{I}$, form the $|\mathcal{I}| \times n$ residual differences, accumulate the row and column contributions to $\boldsymbol{c}$, and sum the associated losses. This evaluates the complete ordered-pair criterion exactly. With $m = \sum_i \Delta_i$ events and block size b, one objective–gradient pass uses $O(nP + mn)$ arithmetic and $O(bn + n + P)$

temporary storage in addition to the stored design. Line-search evaluations contribute further work.

The residual matrix need not be stored either. For any $\boldsymbol{v} = (\boldsymbol{v}_\theta^\top, \boldsymbol{v}_\gamma^\top, \boldsymbol{v}_\beta^\top)^\top$ and $\boldsymbol{c} \in \mathbb{R}^n$, the identities

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{v} &= \boldsymbol{Z}_c\boldsymbol{v}_\theta + \widehat{\boldsymbol{F}}(\boldsymbol{v}_\gamma - \widehat{\boldsymbol{B}}^\top\boldsymbol{v}_\beta) + \boldsymbol{X}_c\boldsymbol{v}_\beta, \\ \widehat{\boldsymbol{W}}^\top\boldsymbol{c} &= \begin{pmatrix} \boldsymbol{Z}_c^\top\boldsymbol{c} \\ \widehat{\boldsymbol{F}}^\top\boldsymbol{c} \\ \boldsymbol{X}_c^\top\boldsymbol{c} - \widehat{\boldsymbol{B}}(\widehat{\boldsymbol{F}}^\top\boldsymbol{c}) \end{pmatrix} \end{aligned} \tag{5.3}$$

allow storage of $\boldsymbol{X}_c$ and the low-rank factors in place of a duplicate residual matrix. Euclidean updates remain in the original $(\boldsymbol{a}, \boldsymbol{\beta})$ coordinates.

5.2 Smoothness and a partially penalized proximal step

Let $\mathscr{L}_\Delta$ be the graph Laplacian with edge weights $\Delta_i + \Delta_j$ for $i < j$. Explicitly,

$$\begin{aligned} \mathscr{L}_\Delta &= n\,\mathrm{diag}(\boldsymbol{\Delta}) + m\boldsymbol{I}_n - \boldsymbol{\Delta}\mathbf{1}_n^\top - \mathbf{1}_n\boldsymbol{\Delta}^\top, \\ \mathscr{B}_h &= \frac{\|\widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}}\|_{\mathrm{op}}}{Nh\sqrt{2\pi}}. \end{aligned} \tag{5.4}$$

Here $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_n)^\top$. Pair grouping and $\phi \le \phi(0)$ imply

$$\begin{aligned} 0 \preceq \nabla^2 f_h(\boldsymbol{\vartheta}) &\preceq \frac{\widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}}}{Nh\sqrt{2\pi}}, \\ \|\nabla f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{y})\|_2 &\le \mathscr{B}_h\|\boldsymbol{x} - \boldsymbol{y}\|_2. \end{aligned} \tag{5.5}$$

This global Lipschitz bound controls numerical step selection. Since $\widehat{\boldsymbol{W}}$ is centered,

$$\begin{aligned} \widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}} &= n\widehat{\boldsymbol{W}}^\top\,\mathrm{diag}(\boldsymbol{\Delta})\widehat{\boldsymbol{W}} + m\widehat{\boldsymbol{W}}^\top\widehat{\boldsymbol{W}}, \\ \mathscr{B}_h &\le \frac{(n+m)\|\widehat{\boldsymbol{W}}\|_{\mathrm{op}}^2}{Nh\sqrt{2\pi}}. \end{aligned}$$

The Frobenius norm gives a computable upper bound in place of the operator norm. Backtracking also permits step selection without evaluating these norms or forming a full Hessian. The factor $1/h$

governs numerical smoothness, separately from the stochastic scale in Section 4.

For $t > 0$, define the soft-thresholding map by $\mathsf{S}_\tau(z)_j = \mathrm{sign}(z_j)(|z_j| - \tau)_+$. The exact proximal-gradient candidate is

$$\begin{aligned} T_t(\boldsymbol{x}) &= \mathrm{prox}_{tg_\lambda}\{\boldsymbol{x} - t\nabla f_h(\boldsymbol{x})\}, \\ [T_t(\boldsymbol{x})]_a &= \boldsymbol{x}_a - t\nabla_a f_h(\boldsymbol{x}), \qquad [T_t(\boldsymbol{x})]_\beta = \mathsf{S}_{t\lambda}\{\boldsymbol{x}_\beta - t\nabla_\beta f_h(\boldsymbol{x})\}. \end{aligned} \tag{5.6}$$

The proximal map leaves the nuisance block unpenalized. Rescaled residual coordinates use the corresponding weighted threshold from Section 3.

5.3 Backtracking algorithm and deterministic convergence

Choose $t_{\max} > 0$ and $\zeta, \omega \in (0, 1)$. From $\boldsymbol{x} = \boldsymbol{x}^{(r)}$, try $t = t_{\max}$ and repeatedly replace t by ζt until, for $\boldsymbol{y} = T_t(\boldsymbol{x})$ and $\boldsymbol{d} = \boldsymbol{y} - \boldsymbol{x}$,

$$f_h(\boldsymbol{y}) \le f_h(\boldsymbol{x}) + \nabla f_h(\boldsymbol{x})^\top \boldsymbol{d} + \frac{\omega}{2t}\|\boldsymbol{d}\|_2^2. \tag{5.7}$$

A nonnegative remainder bound permits stable acceptance without subtracting nearly equal objective values. Write $d_{ij} = \boldsymbol{w}_{ij}^\top(\boldsymbol{y} - \boldsymbol{x})$ and let m_{ij} be the minimum absolute residual difference along the segment from $r_{ij}(\boldsymbol{x})$ to $r_{ij}(\boldsymbol{y})$. Thus $m_{ij} = 0$ when the segment crosses zero and otherwise equals the smaller endpoint magnitude. Taylor's formula gives

$$\begin{aligned} &f_h(\boldsymbol{y}) - f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) \le \mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}), \\ &\mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2Nh}\sum_{i \ne j} \Delta_i \phi(m_{ij}/h) d_{ij}^2 \\ &\qquad\qquad \le \frac{\mathscr{B}_h}{2}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \end{aligned} \tag{5.8}$$

The reference implementation accepts when $\mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}) \le \omega\|\boldsymbol{y} - \boldsymbol{x}\|_2^2/(2t)$. This sufficient test passes for $t \le \omega/\mathscr{B}_h$ and can be evaluated in event-row blocks.

Set $t_r = t$ and $\boldsymbol{x}^{(r+1)} = \boldsymbol{y}$. The cap $t_{\max}$ supplies a uniform step-size bound; $\zeta = \omega = 1/2$ is one admissible numerical choice.

Algorithm 1. Fixed-target, partially penalized proximal gradient
Input: $\boldsymbol{Y}, \boldsymbol{\Delta}, \widehat{\boldsymbol{W}}, q$, fixed $h, \lambda > 0$, finite $\boldsymbol{x}^{(0)}$, $t_{\max} > 0$, and $\zeta, \omega \in (0, 1)$.
Preflight: validate dimensions, scales, event coding, PCA rank, and the nuisance-direction diagnostic in Section 5.5. Return a distinct no-event status when the event set is empty.
For $r = 0, 1, \ldots$: compute $f_h(\boldsymbol{x}^{(r)})$ and $\nabla f_h(\boldsymbol{x}^{(r)})$ by blocks; initialize $t = t_{\max}$; form (5.6) and backtrack until (5.7) is certified by (5.8); accept the candidate; recompute its gradient and record the objective, step size, and full optimality residuals. Record the stopping status and full-coordinate residual.
Output: coefficients in the original scale, frozen prediction map, all tuning values, residuals, iteration and evaluation counts, and diagnostic status. Iteration and line-search limits retain separate status codes.

Theorem 5.1 (Convergence for a fixed, attained objective)**.** *Suppose the finite-sample minimizer set $\mathcal{S} = \arg\min Q$ is nonempty. In exact arithmetic, Algorithm 1 accepts a step after finitely many backtracking trials. If $\mathscr{B}_h > 0$, all accepted steps satisfy*

$$t_r \geq t_{\min} := \min\{t_{\max}, \zeta\omega/\mathscr{B}_h\} > 0. \tag{5.9}$$

When $\mathscr{B}_h = 0$, set $t_{\min} = t_{\max}$. The sequence converges to a member of $\mathcal{S}$, and, for each $\boldsymbol{x}^\star \in \mathcal{S}$ and $r \geq 1$,

$$Q(\boldsymbol{x}^{(r+1)}) \leq Q(\boldsymbol{x}^{(r)}) - \frac{2-\omega}{2t_r}\|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2, \tag{5.10}$$

$$0 \leq Q(\boldsymbol{x}^{(r)}) - Q(\boldsymbol{x}^\star) \leq \frac{\|\boldsymbol{x}^{(0)} - \boldsymbol{x}^\star\|_2^2}{2t_{\min} r}. \tag{5.11}$$

No strong convexity or uniqueness of the minimizer is required.

For fixed n, h, λ, Theorem 5.1 gives an $O(r^{-1})$ objective-gap bound for Algorithm 1 in exact arithmetic. Appendix E specializes standard proximal-gradient reasoning (Beck and Teboulle, 2009; Parikh and Boyd, 2014) to the stated line search. Finite numerical stopping uses the residual below.

5.4 Optimality residuals and the finite-tolerance interface

For the Karush–Kuhn–Tucker (KKT) conditions, at a finite candidate $\boldsymbol{x} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top$, let $\boldsymbol{g} = \nabla f_h(\boldsymbol{x})$ and define

$$\begin{aligned} \mathfrak{r}_{\mathrm{KKT}}(\boldsymbol{x}) &= \max\{\|\boldsymbol{g}_a\|_\infty, \max_j \rho_j(\boldsymbol{x})\}, \\ \rho_j(\boldsymbol{x}) &= \begin{cases} |g_{\beta,j} + \lambda \operatorname{sign}(\beta_j)|, & \beta_j \neq 0, \\ (|g_{\beta,j}| - \lambda)_+, & \beta_j = 0. \end{cases} \end{aligned} \tag{5.12}$$

A maximum over an empty block is zero. This equals $\inf\{\|\boldsymbol{s}\|_\infty : \boldsymbol{s} \in \partial Q(\boldsymbol{x})\}$, and is zero if and only if the finite vector $\boldsymbol{x}$ is a global minimizer. The proximal-gradient mapping

$$G_t(\boldsymbol{x}) = \{\boldsymbol{x} - T_t(\boldsymbol{x})\}/t \tag{5.13}$$

provides a second, continuous fixed-point diagnostic, also zero exactly at a minimizer. At an accepted pair $(\boldsymbol{x}, \boldsymbol{y})$, the explicit vector

$$\begin{gathered} \boldsymbol{s}_y = G_t(\boldsymbol{x}) + \nabla f_h(\boldsymbol{y}) - \nabla f_h(\boldsymbol{x}) \in \partial Q(\boldsymbol{y}), \\ \|\boldsymbol{s}_y\|_2 \leq (1 + t\mathscr{B}_h)\|G_t(\boldsymbol{x})\|_2 \end{gathered} \tag{5.14}$$

links the iteration to a subgradient residual. Equation (5.12) uses the stored coefficient's exact zero/nonzero status.

The implementation reports absolute residuals, the declared tolerance, and the normalized residual $\mathfrak{r}_{\mathrm{KKT}}/\max\{\lambda, \|\boldsymbol{g}\|_\infty\}$. Numerical stationarity and nuisance-direction geometry are assessed separately. Supplementary Appendix G specifies the tracked arithmetic-error budget and the distinction between this numerical diagnostic and an objective-gap certificate.

Corollary 5.1 (Finite optimization tolerance on the statistical event)**.** *Under the assumptions and tuning of the ordinary-model conditional theorem in Section 4, let* $k = s + q$, $\boldsymbol{\vartheta}^\dagger = (\boldsymbol{\theta}_0^\top, (\widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0)^\top, \boldsymbol{\beta}_0^\top)^\top$, *and* $R = 8\sqrt{k}\lambda/\kappa$. *On the same event used in that theorem, every finite* $\widetilde{\boldsymbol{\vartheta}}$ *satisfying*

$$\mathfrak{r}_{\mathrm{KKT}}(\widetilde{\boldsymbol{\vartheta}}) \leq \lambda/4 \tag{5.15}$$

obeys

$$\begin{gathered} \|\widetilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 \leq R, \\ \|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 \leq 4\sqrt{k}R. \end{gathered} \tag{5.16}$$

Thus the conditional probability is still at least $1 - 2\alpha$*, under the same fixed-tuning hypotheses; no extra event is introduced by this deterministic implication.*

A tilted sublevel argument transfers the Section 4 score and contrast bounds to finite candidates under the same fixed-tuning conditions. The experiments use a tighter numerical threshold of 10^{-7}; outcome-based tuning follows the separate validation procedure of Section 3.6.

5.5 Unpenalized escape directions and finite existence

Define the event-anchored nuisance differences $\boldsymbol{a}_{ij} = \widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j$ and the flat-direction space

$$\mathcal{N}_A = \{\boldsymbol{v} \in \mathbb{R}^q : \boldsymbol{a}_{ij}^\top \boldsymbol{v} = 0 \text{ whenever } \Delta_i = 1,\ j \neq i\}.$$

A *strict escape direction* is a vector $\boldsymbol{v}$ such that

$$\begin{aligned} \boldsymbol{a}_{ij}^\top \boldsymbol{v} &\leq 0 \quad (\Delta_i = 1,\ j \neq i), \\ \boldsymbol{a}_{ij}^\top \boldsymbol{v} &< 0 \quad \text{for at least one such pair.} \end{aligned} \tag{5.17}$$

Strictness for just one pair suffices because $\ell_h'(r) > 0$ at every finite r when $h > 0$.

Proposition 5.1 (An exact-arithmetic finite-existence criterion)**.** *For finite* $\boldsymbol{Y}, \widehat{\boldsymbol{W}}$*,* $h, \lambda > 0$*, the set* $\arg\min Q$ *is nonempty if and only if no strict escape direction in* (5.17) *exists. If no such direction exists, every nonempty sublevel set is compact after projecting* $\boldsymbol{a}$ *onto* $\mathcal{N}_A^\perp$*. Directions in* $\mathcal{N}_A$ *leave the objective unchanged and need not imply nonexistence. When* $q = 0$*, a finite minimum always exists.*

Proposition 5.1 characterizes finite attainment of the fixed criterion, including flat nuisance directions. The coefficient guarantees in Section 4 additionally impose restricted-design separation.

If $m > 0$, choose an event index i_0 and let $\mathcal{E} = \{i : \Delta_i = 1\}$, $\mathcal{C} = \{j : \Delta_j = 0\}$. The escape test can be expressed with $O(nq)$ coefficients rather than all pair differences: find $\boldsymbol{v} \in \mathbb{R}^q$ satisfying

$$\begin{aligned} (\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &= 0 \quad (i \in \mathcal{E}), \\ (\widehat{\boldsymbol{A}}_j - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &\geq 0 \quad (j \in \mathcal{C}), \\ \sum_{j \in \mathcal{C}} (\widehat{\boldsymbol{A}}_j - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &\geq 1. \end{aligned} \tag{5.18}$$

The last constraint fixes the scale of a positive strict margin. In exact arithmetic, feasibility is equivalent to (5.17). Numerical output includes equality residuals, inequality margins, and an explicit inconclusive status when the margins cannot be resolved.

For $m = 0$, $Q = \lambda\|\boldsymbol{\beta}\|_1$ and every $\boldsymbol{a}$ with $\boldsymbol{\beta} = 0$ is optimal, but the sample provides no event information. For $m = n$, strict escape is impossible because both directions of every comparison are present. Rank deficiency can still leave flat nuisance coordinates. These cases distinguish nonexistence, nonuniqueness, and lack of event information.

5.6 Regularization paths and stable arithmetic

For fixed (K, h), suppose $\boldsymbol{a}^{(0)}$ attains the minimum of $\boldsymbol{a} \mapsto f_h(\boldsymbol{a}, 0)$. The level

$$\lambda_{\text{zero}} = \|\nabla_\beta f_h(\boldsymbol{a}^{(0)}, 0)\|_\infty \tag{5.19}$$

makes $(\boldsymbol{a}^{(0)}, 0)$ a global minimizer for every $\lambda \geq \lambda_{\text{zero}}$ by the full KKT conditions. The threshold is evaluated after checking nuisance-only stationarity. Penalty paths retain strictly positive levels and use the preceding successful fit as a warm start. This construction follows regularized AFT practice (Suder and Molstad, 2022), with the null fit including the unpenalized block.

For a very negative residual difference, evaluating $r\Phi(r/h) + h\phi(r/h)$ naively subtracts nearly equal quantities. A useful stable identity, with $z = |r|/h$, is

$$\ell_h(r) = r_+ + h\phi(z)J(z), \qquad J(z) = \int_0^\infty u \exp(-zu - u^2/2)\,\mathrm{d}u > 0. \tag{5.20}$$

For $z > 0$, an alternating tail expansion is

$$\begin{aligned} J(z) &= \sum_{j=0}^{M-1} \frac{(-1)^j (2j+1)!!}{z^{2j+2}} + \mathcal{E}_M(z), \\ |\mathcal{E}_M(z)| &\le \frac{(2M+1)!!}{z^{2M+2}}. \end{aligned} \tag{5.21}$$

Here M is the expansion length. The implementation combines normal-tail routines, log-scale evaluation, and the explicit remainder bound. Numerical metadata track tail approximation, unresolved arithmetic, and the line-search budget; details appear in Supplementary Appendix G.

Every candidate retains its declared objective, complete-coordinate stopping rule, and frozen prediction map. Output states distinguish stationarity to tolerance, iteration limits, numerical or line-search failures, nuisance escape, flat directions, and event-free samples.

5.7 Accelerated proximal iterations

The numerical studies use the non-restarted backtracking fast iterative shrinkage-thresholding algorithm (FISTA) (Beck and Teboulle, 2009). Let $\boldsymbol{x}^0 = \boldsymbol{y}^0$ and $\tau_0 = 1$. At iteration r, the smooth gradient is evaluated at the extrapolated point $\boldsymbol{y}^r$. Starting from the previous accepted inverse step L_{r-1}, increase L_r geometrically until the partially penalized candidate

$$\begin{aligned} \boldsymbol{x}_a^{r+1} &= \boldsymbol{y}_a^r - L_r^{-1} \nabla_a f_h(\boldsymbol{y}^r), \\ \boldsymbol{x}_\beta^{r+1} &= \mathsf{S}_{\lambda/L_r}\{\boldsymbol{y}_\beta^r - L_r^{-1} \nabla_\beta f_h(\boldsymbol{y}^r)\} \end{aligned} \tag{5.22}$$

satisfies the full-segment sufficient check

$$\mathfrak{B}_h(\boldsymbol{y}^r, \boldsymbol{x}^{r+1}) \le \frac{\omega L_r}{2} \|\boldsymbol{x}^{r+1} - \boldsymbol{y}^r\|_2^2. \tag{5.23}$$

Here $\mathfrak{B}_h$ is defined in (5.8). The implemented values are $\omega = 1/2$, an initial inverse step of 1, and a doubling factor of 2. The accepted L_r is carried forward and does not decrease. After acceptance,

$$\begin{aligned} \tau_{r+1} &= \frac{1+\sqrt{1+4\tau_r^2}}{2}, \\ \boldsymbol{y}^{r+1} &= \boldsymbol{x}^{r+1} + \frac{\tau_r - 1}{\tau_{r+1}}(\boldsymbol{x}^{r+1} - \boldsymbol{x}^r). \end{aligned} \tag{5.24}$$

The full gradient and KKT residual are recomputed at the returned point $\boldsymbol{x}^{r+1}$. Acceptance controls the quadratic majorant at the extrapolated point $\boldsymbol{y}^r$, so successive accepted objectives can increase. Theorem 5.1 applies to the reference Algorithm 1. The accelerated implementation uses the same objective and full-coordinate finite-tolerance interface to the statistical analysis.

5.8 Qualified active-face Newton refinement

An approximately identified sparse face can be refined using its local curvature. This uses standard second-order ideas for composite optimization (Lee et al., 2014). At a finite input point, define

$$\mathcal{A} = \{1, \ldots, q\} \cup \{j > q : \vartheta_j \neq 0\}, \qquad Z = \{j > q : \vartheta_j = 0\}.$$

Retain this set and the signs of the penalized active coordinates during the local call. Put $v_j = 0$ for the unpenalized coordinates and $v_j = \operatorname{sign}(\vartheta_j)$ for the nonzero penalized coordinates. With $\boldsymbol{g} = \nabla f_h(\boldsymbol{\vartheta})$ and $\boldsymbol{H} = \nabla^2 f_h(\boldsymbol{\vartheta})$, solve

$$\begin{aligned} \boldsymbol{H}_{\mathcal{A}\mathcal{A}}\boldsymbol{d}_{\mathcal{A}} &= -(\boldsymbol{g}_{\mathcal{A}} + \lambda \boldsymbol{v}_{\mathcal{A}}), \\ \boldsymbol{d}_Z &= 0. \end{aligned} \tag{5.25}$$

The numerically positive-definite active system is solved directly by Cholesky factorization.

Let $\widehat{s}_{\mathcal{A}} = \widehat{\boldsymbol{g}}_{\mathcal{A}} + \lambda \boldsymbol{v}_{\mathcal{A}}$ and let E denote the tracked gradient error budget. For a trial $t > 0$ inside the original sign face, use the sufficient Armijo check

$$\begin{aligned} t\widehat{s}_{\mathcal{A}}^{\top}\boldsymbol{d}_{\mathcal{A}} + \mathfrak{B}_h(\boldsymbol{\vartheta}, \boldsymbol{\vartheta} + t\boldsymbol{d}) + tE\|\boldsymbol{d}\|_1 &\leq c_A t\widehat{s}_{\mathcal{A}}^{\top}\boldsymbol{d}_{\mathcal{A}}, \\ c_A &= 10^{-4}. \end{aligned} \tag{5.26}$$

Trial steps are halved as necessary. A 0.99 fraction-to-boundary rule retains the input signs, and every accepted candidate is checked on all original coordinates, including the numerical condition $|\widehat{g}_j| + E \leq \lambda$ for every $j \in Z$. Here E is the candidate's tracked gradient error budget. The local call stops if the fixed face, numerical curvature, or line-search requirements cease to hold. Passing the complete KKT test is required for a successful output.

5.9 Common numerical policy for training comparisons

Both estimators use the same sequence of numerical operations in the training-validation study. Each candidate first receives up to 30,000 accepted FISTA updates at full-coordinate tolerance 10^{-7}. A successful return is used directly. A return at the update limit qualifies for at most one active-face refinement if its last five distinct saved accepted checkpoints have identical exact $\boldsymbol{\beta}$ support and signs, the gradient budget is at most 10^{-8}, and all zero-coordinate margins remain strictly positive after allowing for that budget. The refinement allows at most ten accepted Newton updates. Otherwise the candidate is recorded as numerically ineligible under the specified budget.

The final parameter vector defines all coefficient blocks, the deployment slope (3.17), fitted scores, and full KKT residual. After refinement, these quantities are reconstructed from the new coefficients and unchanged design. The preceding successful candidate supplies the next warm start. This policy and the selection rule jointly define the procedure evaluated in Section 6.4.

6 Computational illustrations and training selection

The numerical study addresses two questions: how factor coordinates relate to computational geometry, and how training selection allocates fitted effects between common and sparse residual components. Six synthetic samples, one per scenario, provide fixed inputs for these illustrations. Curvature and stationarity diagnostics address the first question; three-fold validation and final-model summaries address the second.

6.1 Synthetic design and comparison protocol

Each cohort has $n = 120$ training patients, $p = 300$ molecular covariates, and $d = 2$ clinical covariates. Independently generate the coordinates of $\boldsymbol{f}_i \in \mathbb{R}^2$, $\boldsymbol{v}_i \in \mathbb{R}^p$, and $\boldsymbol{Z}_i \in \mathbb{R}^2$ from $\mathrm{Unif}[-\sqrt{3}, \sqrt{3}]$. For $j = 1, \ldots, p$, let

$$\begin{aligned} \boldsymbol{B}_{j\cdot} &= \sqrt{\rho}\left(\cos\frac{2\pi(j-1)}{p},\ \sin\frac{2\pi(j-1)}{p}\right), \\ \boldsymbol{u}_i &= \sqrt{1-\rho}\,\boldsymbol{v}_i, \qquad \boldsymbol{X}_i = \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i. \end{aligned} \tag{6.1}$$

The ordinary-model signal is $\eta_i = \boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top \boldsymbol{\beta}_0$, with $\boldsymbol{\theta}_0 = (0.35, -0.25)^\top$ and nonzero $\boldsymbol{\beta}_0$ entries $(0.6, -0.6, 0.5, -0.5, 0.4)$ at indices $(1, 61, 121, 181, 241)$. All remaining entries are zero. The error is standard normal, $a_3 t_3$, or $a_C C$ with C standard Cauchy, where

$$a_3 = \Phi^{-1}(0.75)/F_{t_3}^{-1}(0.75), \qquad a_C = \Phi^{-1}(0.75).$$

Thus the error interquartile ranges (IQRs) agree. Independently draw $V_i \sim N(0, 1)$ and set

$$\begin{aligned} \log T_i &= \eta_i + \varepsilon_i, \\ \log C_i &= \eta_i + c + V_i, \\ \int F_\varepsilon(c+v)\phi(v)\,\mathrm{d}v &= 1 - \pi_C. \end{aligned} \tag{6.2}$$

Equation (6.2) calibrates the population censoring proportion π_C. Realized event indicators follow the generated event and censoring times, retaining sample variation in censoring. Computation remains on the log-time scale.

The factor-adjusted estimator (FA) uses $K = 2$ and the unadjusted estimator (UA) uses $K = 0$; the loss, molecular penalty, and bandwidth $h = 0.2$ agree. S1 is the prespecified over-adjustment control, with $\rho = 0$. At the fixed coefficients, signal variance is approximately 1.565, 1.3254, and 0.8462 for $\rho = 0, 0.25, 0.75$. The scenarios therefore vary dependence at fixed coefficients, with the corresponding change in signal variance.

Table 1 Synthetic-design scenarios. The error scales match interquartile ranges

Scenario	ρ	Target censoring	Error distribution
S1	0	30%	Standard normal
S2	0.25	30%	Standard normal
S3	0.75	30%	Standard normal
S4	0.75	30%	IQR-scaled t_3
S5	0.75	60%	IQR-scaled t_3
S6	0.75	30%	IQR-scaled Cauchy

6.2 Active curvature and work to common accuracy

Both estimators use non-restarted FISTA at $\lambda = 0.01$, initialized from their respective $\lambda = 0.025$ fits on S3, S4, and S6. Each method retains its original parameter coordinates and training design. The common stopping threshold is 10^{-7} for the complete KKT residual plus the tracked numerical budget.

Factor adjustment reaches the common accuracy levels with fewer gradient evaluations in these three samples (Figure 1). At 10^{-6}, first-crossing counts for FA and UA are 2,549 and 33,195 in S3, and 3,021 and 34,693 in S4. S6 FA reaches this level at 4,843 evaluations; S6 UA remains above it at its endpoint. At the prescribed 10^{-7} stopping level, FA takes 3,619, 3,271, and 4,507 accepted updates. UA reaches its 30,000-update limit with residuals 4.269×10^{-7}, 3.033×10^{-7}, and 2.686×10^{-6}. The accepted-update view is provided in Supplementary Appendix F.

For each saved endpoint, form the active set from all unpenalized coordinates and the exactly nonzero molecular coefficients. The local loss Hessian is

$$\boldsymbol{H}(\boldsymbol{\vartheta}) = \frac{1}{Nh} \sum_{i \neq j} \Delta_i \phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \, \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^{\top}. \tag{6.3}$$

We analyze $\boldsymbol{H}_{\mathcal{A}\mathcal{A}}$ using the numerical rank threshold

$$100 \epsilon_{\text{mach}} |\mathcal{A}| \max_j |\lambda_j(\boldsymbol{H}_{\mathcal{A}\mathcal{A}})|.$$

All six active Hessians were numerically full rank under this rule.

The spectral differences are driven primarily by the upper end: FA reduces the maximum active curvature from 89–116 to 2.89–3.70, with smaller active spectral ratios in all three samples (Table 2).

Table 2 Active loss-Hessian spectra at the FISTA endpoints. Ratios use the smallest numerical positive eigenvalue

Scenario	Method	Dimension	Minimum	Maximum	Ratio
S3	FA	103	9.055×10^{-4}	2.893	3.195×10^{3}
S3	UA	101	1.055×10^{-3}	89.074	8.445×10^{4}
S4	FA	101	2.188×10^{-3}	2.888	1.320×10^{3}
S4	UA	100	1.508×10^{-3}	94.353	6.256×10^{4}
S6	FA	102	4.942×10^{-4}	3.704	7.494×10^{3}
S6	UA	104	2.463×10^{-4}	115.830	4.702×10^{5}

Minimum curvature varies by scenario and is slightly lower for FA in S3. The corresponding final scalar steps are 1/4 for FA and 1/256 for UA. UA's inverse step remains $L = 256$ throughout its last 5,000 updates, with no further backtracking. The lower upper-curvature scale is therefore consistent with the larger scalar steps used by FA.

The largest endpoint residual lies in a nonzero molecular coefficient for all six fits. Exactly zero coordinates satisfy the KKT inequalities at the endpoints and selected late checkpoints. Across UA updates 26,000, 27,000, 28,000, 29,000, and 30,000, the saved supports and signs agree while the dominant residual coordinate changes. Clinical-block residuals also exceed 10^{-7} for S3 and S6 UA, confirming the need for full-coordinate accuracy.

Weak-curvature directions explain where the remaining stationarity error lies. The ten smallest-eigenvalue directions contain 79.1%, 80.3%, and 52.8% of the squared active residual for S3, S4, and S6 UA. FA residuals also concentrate in weak directions. For UA, this concentration combines with a wider active spectrum and a fixed scalar step, producing small late objective ranges alongside oscillating KKT residuals. These comparisons use each method's own endpoint, active set, and optimization domain.

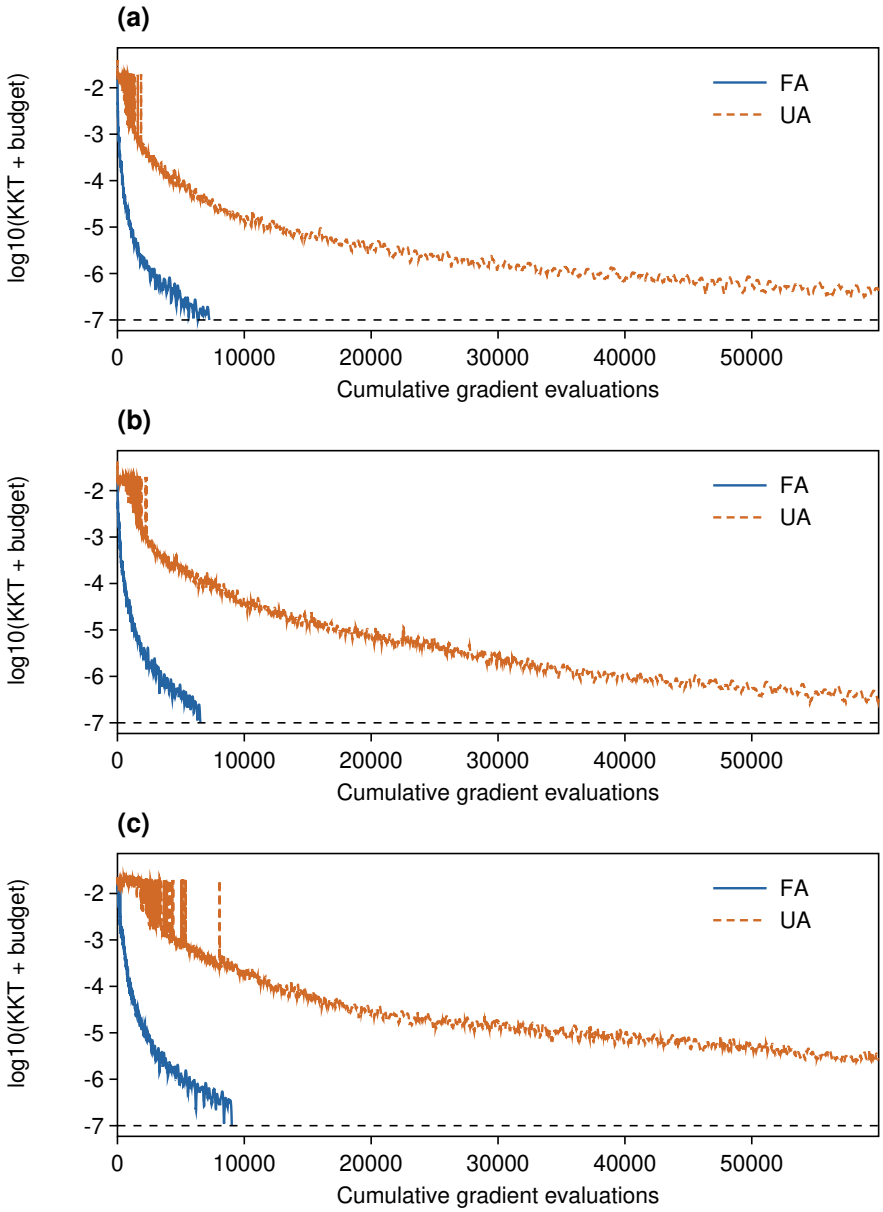


Fig. 1 Computational work to a common stationarity tolerance. The full original-coordinate KKT residual plus its tracked numerical budget is plotted against cumulative gradient evaluations for S3, S4, and S6 in panels (a), (b), and (c). FA is solid and UA is dashed. Both methods use non-restarted FISTA, $h = 0.2$, $\lambda = 0.01$, and their respective $\lambda = 0.025$ starts. The horizontal line is 10^{-7}. Curves end at their FISTA stopping points, before the local refinement in Section 6.3

6.3 Active-face refinement resolves late-stage residuals

One active-face Newton step brings each UA endpoint below 10^{-7} while preserving its design, support, signs, $h = 0.2$, and $\lambda = 0.01$ (Table 3). All three FA endpoints meet the same accuracy threshold before refinement and return unchanged.

Table 3 Full original-coordinate KKT residuals for one-step refinement of the UA fits

Scenario	Before	After	Newton updates
S3	4.269×10^{-7}	4.864×10^{-10}	1
S4	3.033×10^{-7}	1.780×10^{-10}	1
S6	2.686×10^{-6}	6.997×10^{-9}	1

All three step lengths were one, with no backtracking halvings. The full steps preserved every active molecular sign and left all original zero coefficients at zero. The new minimum inactive-coordinate margins were approximately 7.75×10^{-5}, 2.76×10^{-6}, and 9.99×10^{-6}. The objective decreases were approximately 2.58×10^{-10}, 1.02×10^{-10}, and 4.94×10^{-9}. Thus substantial reductions in full stationarity residual were obtained without changing the support or statistical objective.

The full steps provide a direct computational counterpart to the spectral diagnosis: local direction-specific scaling completes stationarity on each existing active face. The training-selection procedure below offers the same refinement rule to both estimators.

6.4 Training selection and factor-mediated prediction

Each of the six 120-patient training cohorts was split once into three event-stratified folds of 40 patients; both methods used the same labels. For every fold and method, centering and PCA were learned on the 80 training patients, and validation predictions used only the resulting frozen transformation. The candidate grid was the fixed absolute set $\{0.15, 0.06, 0.025, 0.01\}$ at $h = 0.2$. The validation score was (3.18), with the zero-prediction contrast computed within each ordered pair and aggregated over the three within-fold pair sets.

Each method selected among candidates valid in all three folds. One invalid fold made the aggregate unavailable. Candidates within $10^{-10} + 10^{-10}\max(1, |\mathrm{CV}_{\min}|)$ of the best valid score were tied, with larger λ preferred. After selection, each full-training refit used freshly estimated preprocessing and a new nuisance initialization.

The procedure produced 143 successful fold candidates out of 144 attempted positions, giving 47 eligible three-fold aggregates out of 48. All 12 selected full-training refits met the full-coordinate KKT-plus-budget threshold of 10^{-7}. Table 4 reports the selections and refit residuals. Figure 2 shows all four penalty positions for each method, marking the selected points and the invalid position.

Table 4 Training-only penalty selections and full-training refit residuals. A star denotes the upper grid boundary

	FA			UA		
Scenario	Eligible	λ	KKT + budget	Eligible	λ	KKT + budget
S1	4/4	0.06	9.825×10^{-8}	4/4	0.06	7.786×10^{-8}
S2	4/4	0.06	9.600×10^{-8}	4/4	0.06	8.981×10^{-8}
S3	4/4	0.06	9.886×10^{-8}	4/4	0.06	9.950×10^{-8}
S4	4/4	0.15*	9.323×10^{-10}	3/4	0.06	9.870×10^{-8}
S5	4/4	0.06	9.503×10^{-10}	4/4	0.025	9.925×10^{-8}
S6	4/4	0.15*	7.446×10^{-10}	4/4	0.15*	9.298×10^{-8}

Both methods select $\lambda = 0.06$ in S1–S3 and 0.15 in S6, with different choices in S4 and S5. Three selections are at the upper grid boundary. S4 fold 1 UA at $\lambda = 0.01$ reaches the FISTA budget without satisfying the support-and-sign refinement gate, so that candidate has no aggregate score. At the selected penalties, FA has a lower validation score in S2, and UA has a lower score in S1 and S3–S6 (Table 5). The scores describe the stated training partitions.

The selected models reveal a structural distinction between residual sparsity and molecular prediction. In S4–S6, FA retains factor-mediated prediction with $\widehat{\boldsymbol{\beta}} = 0$; its deployment slope has 300 exactly nonzero entries in every scenario. UA's deployment slope equals its molecular coefficient vector and shares its nonzero count (Table 5). These quantities summarize different roles in the fitted model: sparse residual effects and total molecular information in the frozen score.

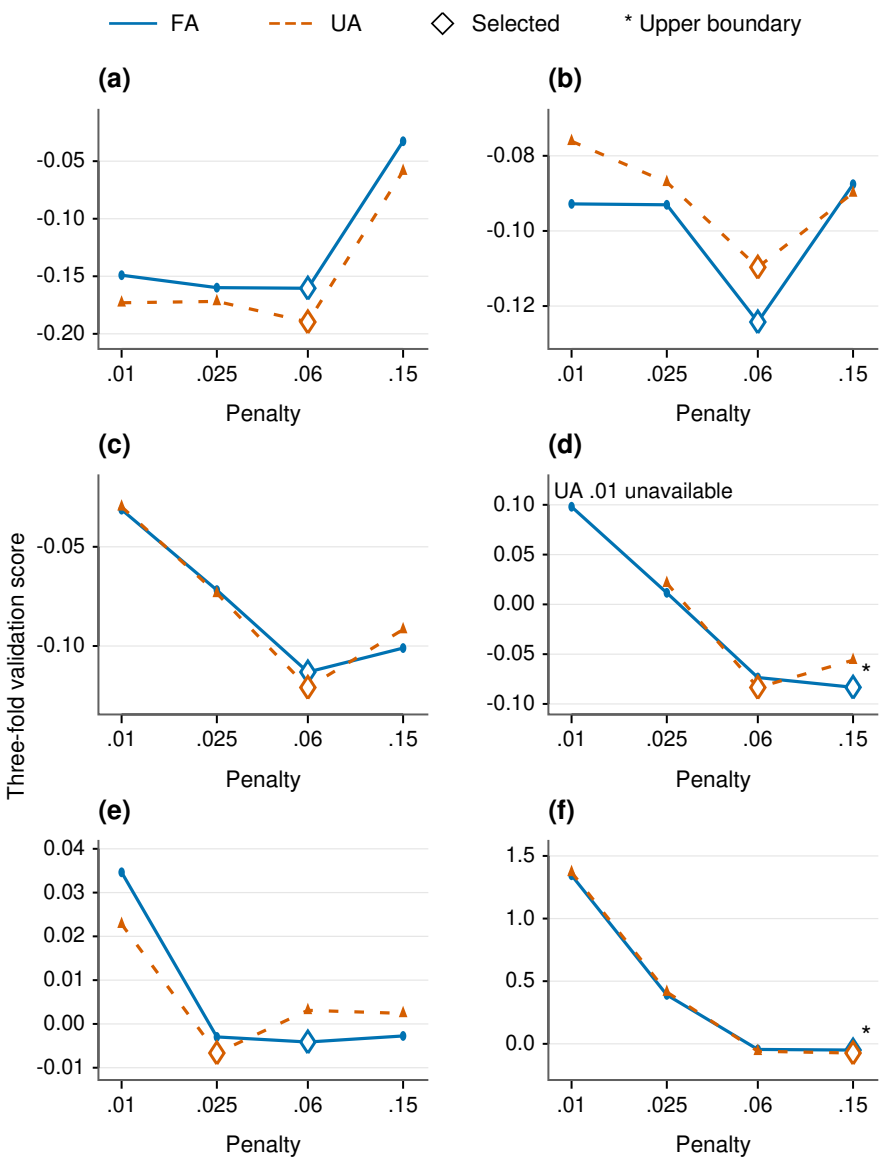


Fig. 2 Training-only three-fold validation paths on S1–S6 in panels (a)–(f). FA uses solid lines and circles; UA uses dashed lines and triangles. Each score is the aggregate within-fold Gehan loss contrast against zero prediction; lower values are preferred. Diamonds mark selected penalties and asterisks mark the upper boundary of the original fixed grid. The S4 UA candidate at $\lambda = 0.01$ has no aggregate score because one fold failed its numerical eligibility rule. Each panel uses one three-fold partition and its own vertical scale

Table 5 Selected full-training models. $\|\hat{\boldsymbol{\beta}}\|_0$ counts exact nonzero penalized coefficients; $\|\hat{\boldsymbol{b}}_{\mathrm{pred}}\|_0$ counts exact nonzero frozen prediction slopes. CV is the three-fold score at the selected penalty; smaller is better. FISTA and Newton columns count accepted updates in the full-training refit

Scenario	Method	CV	$\|\hat{\boldsymbol{\beta}}\|_0$	$\|\hat{\boldsymbol{b}}_{\mathrm{pred}}\|_0$	FISTA	Newton
S1	FA	-0.160420	58	300	449	0
S1	UA	-0.189698	55	55	314	0
S2	FA	-0.124238	36	300	209	0
S2	UA	-0.109671	38	38	1439	0
S3	FA	-0.113124	6	300	145	0
S3	UA	-0.120923	11	11	9178	0
S4	FA	-0.083292	0	300	0	0
S4	UA	-0.083487	7	7	7184	0
S5	FA	-0.004137	0	300	0	0
S5	UA	-0.006646	23	23	12054	0
S6	FA	-0.049824	0	300	0	0
S6	UA	-0.073506	2	2	1755	0

Nine UA fold candidates reach the prescribed accuracy after one qualified Newton update each. All twelve selected full-training fits finish without refinement. The S4–S6 FA refits satisfy the stopping test at their new nuisance initializations and take zero FISTA updates. Table 5 reports refit work; Supplementary Appendix F includes initialization and complete work accounting.

7 Discussion

The finite-sample analysis connects same-sample factor learning to estimation of the original sparse AFT slope. The reference representation preserves the ordinary-model predictor exactly, placing the effect of factor estimation in the design geometry. Conditional score centering, column-scale empirical-process control, and local event information then yield coefficient bounds and finite attainment under explicit assumptions. Additional factor effects enter through a separate approximation term relative to the specified latent representation.

This separation is also useful computationally. In the fixed strong-dependence illustrations, FA has lower maximum active curvature and smaller active spectral ratios, together with larger scalar steps

and less work to common accuracy. Late UA residuals concentrate in weak-curvature directions on the selected stable supports. A single active-face Newton correction supplies the direction-specific scaling needed to complete each of those fits. The numerical policy combines proximal search with qualified local refinement, using the same full-coordinate accuracy criterion for both estimators.

Training selection gives complementary information about the representation. Some FA fits retain a dense factor-mediated molecular score with a zero residual coefficient block. Thus $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{b}}_{\mathrm{pred}}$ answer different questions: the former describes residual associations, while the latter records the molecular slope deployed for prediction. The six training partitions exhibit different selected-score orderings across methods, making both coefficient recovery and independent prediction relevant to evaluation of the complete selection procedure.

The theoretical bounds concern fixed or covariate-measurable tuning, and the numerical experiments describe fixed-sample geometry and training selection. Independent repeated samples can assess the risk of the outcome-selected procedure under prespecified grids and numerical rules. A biomedical application would additionally require a common endpoint, an explicit observation scheme, and evaluation of the frozen prediction map on an external cohort. These evaluations extend the present theory and computational illustrations to statistical performance and applied use.

The resulting framework links three objects: the original sparse AFT target, covariate geometry learned from the sample, and the numerical accuracy of the fitted estimator. Exact reference coordinates preserve the target, finite-sample analysis accounts for censoring and factor estimation, and local curvature completes the computational connection. Together, they provide a coherent basis for sparse censored regression with estimated factor structure.

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Ethics approval The work develops statistical theory and uses synthetic data. No human participants, animals, or identifiable personal data were studied.

Consent to participate and consent for publication Not applicable: the study contains no human-participant data or identifiable personal information.

Data and code availability Data-generating mechanisms and numerical procedures are described in the article and Online Resource 1. [Author confirmation required: supply an accessible repository and persistent identifier, or authorize an accurate availability-on-request statement for the synthetic data and implementation.]

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Supplementary information Online Resource 1 contains proofs, numerical evidence descriptions, implementation details, and the supplementary accepted-update trajectory figure.

A Proof of Proposition 2.1

We give the proof for a fixed latent representation. The observed-design statement follows by the same argument, as explained at the end. All conditional expectations in the first three steps are given $(\boldsymbol{D}_1, \boldsymbol{D}_2)$. Write $G_i(t) = G(t \mid \boldsymbol{D}_i)$, $\boldsymbol{w} = \boldsymbol{D}_1 - \boldsymbol{D}_2$, and $\boldsymbol{v} = \boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0$.

Event-weighted differences. On $\{\Delta_i = 1\}$, $R_i = \varepsilon_i$, so the conditional event subdensity is $f_\varepsilon(t)G_i(t)$. Moreover,

$$\Pr(R_j \geq t \mid \boldsymbol{D}_j) = S_\varepsilon(t)G_j(t).$$

It follows that

$$\begin{aligned} A_{12} &:= E\{\Delta_1 \mathbf{1}(R_2 \geq R_1) \mid \boldsymbol{D}_1, \boldsymbol{D}_2\} \\ &= \int f_\varepsilon(t)S_\varepsilon(t)G_1(t)G_2(t)\,\mathrm{d}t = A_{21}. \end{aligned} \tag{A.1}$$

The finite measure $\Pr(\Delta_1 = 1, R_2 - R_1 \in \mathrm{d}r \mid \boldsymbol{D}_1, \boldsymbol{D}_2)$ has a density, say $q_{12}(r)$. With $F_{R_2|\boldsymbol{D}_2}$ denoting the conditional distribution of R_2,

$$q_{12}(r) = \int f_\varepsilon(z - r)G_1(z - r)\,\mathrm{d}F_{R_2|\boldsymbol{D}_2}(z) \leq L_\varepsilon \qquad \text{a.e.} \tag{A.2}$$

The Stieltjes integral allows censoring-induced atoms in R_2. Restricting it to the event submeasure of patient 2 gives, for $|r| \leq A$,

$$\begin{aligned} q_{12}(r) &\geq \int f_\varepsilon(z - r)G_1(z - r)f_\varepsilon(z)G_2(z)\,\mathrm{d}z \\ &\geq (2A - |r|)m_\varepsilon^2 g_0^2 \geq c_{\mathrm{loc}}, \end{aligned} \tag{A.3}$$

because $I \cap (I + r)$ has length $2A - |r|$. These bounds hold almost everywhere, which suffices for the integrations below.

A finite conditional contrast. Set $t = \boldsymbol{w}^\top \boldsymbol{v}$. At the candidate parameter,

$$e_2^o(\boldsymbol{\vartheta}) - e_1^o(\boldsymbol{\vartheta}) = R_2 - R_1 + t.$$

Define

$$k_{12}(t) = E[\Delta_1\{(R_2 - R_1 + t)_+ - (R_2 - R_1)_+\} \mid \boldsymbol{D}_1, \boldsymbol{D}_2]\,.$$

This is finite, even without an error moment, because its absolute value is at most $|t|$. The absolute continuity of the event-weighted

difference measure gives $k'_{12}(0) = A_{12}$ and $k''_{12}(t) = q_{12}(-t)$ almost everywhere. Integrating this derivative, for $t \geq 0$,

$$k_{12}(t) - A_{12}t = \int_0^t (t - z) q_{12}(-z)\,\mathrm{d}z.$$

For $t < 0$, the corresponding expression is

$$k_{12}(t) - A_{12}t = \int_t^0 (z - t) q_{12}(-z)\,\mathrm{d}z.$$

Nonnegativity of the density and (A.3) imply in either case

$$k_{12}(t) - A_{12}t \geq \frac{c_{\mathrm{loc}}}{2}\rho_A(t). \tag{A.4}$$

For example, when $t > A$ the retained integral is $c_{\mathrm{loc}} \int_0^A (t - z)\,\mathrm{d}z = c_{\mathrm{loc}}(At - A^2/2)$; the case $0 \leq t \leq A$ gives $c_{\mathrm{loc}}t^2/2$. The negative cases follow analogously.

Cancellation and strict separation. Since the two patients are identically distributed, interchanging their indices preserves their joint law, changes $\boldsymbol{w}$ to $-\boldsymbol{w}$, and leaves A_{12} unchanged by (A.1). The integrability of $\boldsymbol{w}^\top \boldsymbol{v}$ therefore gives

$$E(A_{12}\boldsymbol{w}^\top \boldsymbol{v}) = 0.$$

Taking expectations in (A.4) proves (2.15). When $\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0 \in \Theta_s$, their difference obeys $\|\boldsymbol{v}_\beta\|_0 \leq 2s$. Assumption 2.3 yields

$$E\{(\boldsymbol{w}^\top \boldsymbol{v})^2\} = 2\,\mathrm{Var}(\boldsymbol{D}_i^\top \boldsymbol{v}) \geq 2\kappa_{\mathrm{pop}}\|\boldsymbol{v}\|_2^2.$$

For nonzero $\boldsymbol{v}$, $\Pr(\boldsymbol{w}^\top \boldsymbol{v} \neq 0) > 0$. Because ρ_A vanishes only at zero, the contrast is strictly positive. Also $\rho_A(t) \leq 2A|t|$, so the expectation in (2.15) is finite. This proves the unique minimum at $\boldsymbol{\vartheta}_0$ on Θ_s without requiring a finite expected uncentered loss.

The ordinary observed-data target. For $\mathcal{M}_0$, set $\boldsymbol{V}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{X}_i^\top)^\top$ and $\boldsymbol{b} = (\boldsymbol{\theta}^\top, \boldsymbol{\beta}^\top)^\top$. Replace $\boldsymbol{w}^\top \boldsymbol{v}$ by $(V_1 - V_2)^\top(\boldsymbol{b} - \boldsymbol{b}_0)$ in the preceding calculation. The event-weighted centering and density bounds can still be established conditional on $(\boldsymbol{D}_1, \boldsymbol{D}_2)$ and then averaged. A positive variance for every nonzero sparse difference in this observed predictor gives strict separation.

Every contrast increment can now be computed as an expectation of a function of the observed pair (O_1, O_2). To avoid an unknown reference parameter in the identification argument, one can fix any deterministic $\boldsymbol{b}_\dagger$ and subtract the loss at $\boldsymbol{b}_\dagger$ instead. Every such contrast is finite and differs from the one centered at $\boldsymbol{b}_0$ by a constant independent of $\boldsymbol{b}$. Thus its unique minimizer is a functional of the observed-data law. The corresponding $\mathcal{M}_1$ result is relative to the specified latent representation. □

B Deterministic properties of the estimator

B.1 Proof of Proposition 3.1

Direct integration of $E(r + hV)_+$ gives (3.6). Jensen's inequality yields $\ell_h(r) \geq r_+$. For $r \geq 0$, the approximation gap $g_h(r) = \ell_h(r) - r$ has derivative $\Phi(r/h) - 1 \leq 0$. The identity $\ell_h(-r) = \ell_h(r) - r$ shows that the gap is even. Its maximum is therefore attained at zero and equals $h\phi(0) = h/\sqrt{2\pi}$. This proves the first inequality in (3.8) without any assumption on observed outcomes.

Symmetry and Fubini's theorem give

$$\begin{aligned}\int_{\mathbb{R}} |\Phi(r/h) - \mathbf{1}\{r \geq 0\}| \, \mathrm{d}r &= 2h \int_0^\infty \{1 - \Phi(t)\} \, \mathrm{d}t \\ &= 2hE(V_+) = h\sqrt{2/\pi}.\end{aligned}$$

The value of the indicator at zero is irrelevant to this integral. Summing the uniform scalar bound uses

$$\frac{1}{n(n-1)} \sum_{i \neq j} \Delta_i = \frac{1}{n} \sum_i \Delta_i = \bar{\Delta},$$

which proves (3.9). Including diagonal terms would add the parameter-independent quantity $n^{-2}h\phi(0) \sum_i \Delta_i$ if the n^{-2} normalization were used, so the stated off-diagonal convention must be retained when comparing objective values.

Differentiating the finite sum gives (3.19), whose Hessian is positive semidefinite. Grouping the coefficients of each $\widehat{\boldsymbol{W}}_i$ proves (3.21); convex subgradient optimality gives (3.20). Differentiation applies directly to the finite empirical sum. Dense gradient evaluation costs $O(nP + n^2)$ arithmetic; pair weights may be blocked to avoid an $N \times P$ matrix.

B.2 Proof of Proposition 3.2 and coordinate identities

The PCA equations imply

$$\boldsymbol{X}_c \boldsymbol{X}_c^\top \widehat{\boldsymbol{F}}/n = \widehat{\boldsymbol{F}} \widehat{\boldsymbol{\Lambda}}, \quad \widehat{\boldsymbol{B}}^\top \widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}^\top \boldsymbol{X}_c \boldsymbol{X}_c^\top \widehat{\boldsymbol{F}}/n^2 = \widehat{\boldsymbol{\Lambda}}.$$

Since $\boldsymbol{X}_c \boldsymbol{X}_c^\top \mathbf{1}_n = 0$, selected positive-eigenvalue eigenvectors are centered. The definitions then give $\boldsymbol{J}\widehat{\boldsymbol{U}} = \widehat{\boldsymbol{U}}$, $\widehat{\boldsymbol{F}}^\top \widehat{\boldsymbol{U}} = 0$, and $\boldsymbol{X}_c \widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}} \widehat{\boldsymbol{\Lambda}}$, proving (3.2).

Substituting (3.13) into the fitted predictor yields

$$\begin{aligned}\widehat{\boldsymbol{W}}\boldsymbol{\vartheta}^{\dagger} &= \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \widehat{\boldsymbol{U}}\boldsymbol{\beta}_0 + \widehat{\boldsymbol{F}}(\widehat{\boldsymbol{B}}^{\top}\boldsymbol{\beta}_0 + \boldsymbol{H}^{-T}\boldsymbol{\xi}_0)\\ &= \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \boldsymbol{X}_c\boldsymbol{\beta}_0 + (\boldsymbol{F}_c\boldsymbol{H}^{\top} + \boldsymbol{E}_F)\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\\ &= \boldsymbol{\eta}_c^0 + \boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{\xi}_0.\end{aligned}$$

Since $\boldsymbol{J}\boldsymbol{E}_F = \boldsymbol{E}_F$, $\mathbf{d}$ is centered. Subtracting from $Y_i = R_i + \eta_i^0$ gives the residual identities; rowwise Cauchy–Schwarz bounds b_n. Adding and subtracting the reference predictor gives the last decomposition. When $\boldsymbol{\xi}_0 = 0$, the approximation term vanishes without any factor-accuracy assumption.

For orthogonal Q, rotating $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{B}}$ to $\widehat{\boldsymbol{F}}\boldsymbol{Q}^{\top}$ and $\widehat{\boldsymbol{B}}\boldsymbol{Q}^{\top}$ preserves $\widehat{\boldsymbol{U}}$; taking $\boldsymbol{\gamma}^* = \boldsymbol{Q}\boldsymbol{\gamma}$ preserves the objective. Writing the prediction map as $(\widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{B}})^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_c$ shows that it rotates to $Q\widehat{\boldsymbol{f}}(\boldsymbol{x})$ and preserves the deployed score, even outside the diagonal PCA eigenbasis.

For a training row $\boldsymbol{x}_{c,i}^{\top}$, the identity $\boldsymbol{X}_c\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}$ gives $\widehat{\boldsymbol{\Lambda}}^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_{c,i} = \widehat{\boldsymbol{f}}_i$. The residual mapping therefore reproduces $\widehat{\boldsymbol{u}}_i$. Finally,

$$\widehat{\boldsymbol{u}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{f}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\gamma}} = \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{\beta}} + \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}(\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{\beta}}),$$

which proves (3.17). This proves exact reproduction of the fitted training factors, residuals, and scores.

B.3 Finite attainment and a boundary example

Assume $q > 0$, $h > 0$, $\lambda > 0$, finite observations, and (3.12). On a nonempty sublevel set $Q_{n,h,\lambda}(\boldsymbol{a}, \boldsymbol{\beta}) \le c$, nonnegativity of ℓ_h gives $\|\boldsymbol{\beta}\|_1 \le c/\lambda$. Write

$$t_{ij} = (\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j)^{\top}\boldsymbol{a}, \qquad z_{ij} = Y_j - Y_i + (\widehat{\boldsymbol{u}}_i - \widehat{\boldsymbol{u}}_j)^{\top}\boldsymbol{\beta}.$$

The inequalities $\ell_h(t + z) \ge (t + z)_+ \ge t_+ - |z|$ imply

$$L_{n,h}(\boldsymbol{a}, \boldsymbol{\beta}) \ge \chi_n\|\boldsymbol{a}\|_2 - C_Y - C_U\|\boldsymbol{\beta}\|_1,$$

where the finite sample constants can be taken as

$$C_Y = N^{-1}\sum_{i\neq j}\Delta_i|Y_j - Y_i|, \qquad C_U = N^{-1}\sum_{i\neq j}\Delta_i\|\widehat{\boldsymbol{u}}_i - \widehat{\boldsymbol{u}}_j\|_{\infty}.$$

Therefore $\|\boldsymbol{a}\|_2 \le \{c + C_Y + C_U c/\lambda\}/\chi_n$ on the sublevel set. It is closed and bounded and hence compact. Taking $c = Q_{n,h,\lambda}(0,0)$ gives a nonempty compact set containing at least one global minimum; the minimizer set is itself compact. If $q = 0$, the penalty directly bounds all coordinates and the same conclusion follows without χ_n.

The sum defining χ_n is nonnegative and continuous on the unit sphere. Thus $\chi_n > 0$ precisely when every nonzero nuisance direction admits a positive event-anchored comparison. The condition concerns the realized event-anchored nuisance design.

For an example of nonattainment, take $n = 2$, $K = 0$, one centered unpenalized covariate with rows -1 and 1, no effective molecular contribution, and $\Delta_1 = 1$, $\Delta_2 = 0$. With scalar coefficient a and $y = Y_2 - Y_1$, the objective reduces to

$$Q(a) = \tfrac{1}{2}\ell_h(y - 2a).$$

It is positive for every finite a, but tends to zero as $a \to \infty$. Thus its infimum is not attained. This remains a valid finite-sample failure mechanism when molecular coefficients are available but optimally zero. Conversely, failure of (3.12) need not imply nonexistence: if all $\Delta_i = 0$, any unpenalized a with $\boldsymbol{\beta} = 0$ is a minimizer, although no slope is learned. Existence, uniqueness, and information for estimation are different requirements.

C Proofs for the conditional theory

Throughout this appendix, E_D and $\Pr_D$ denote expectation and probability conditional on D_n. The design, $\boldsymbol{\vartheta}^\dagger$, h, λ, and the approximation vector $\mathbf{d}$ are fixed. Independent ghost outcomes used below have each patient's own conditional law. The conditional laws are retained separately for each patient.

C.1 Event-weighted differences and the mean score

For each ordered pair let $Q_{ij} = R_j - R_i$, $\zeta_{ij} = d_i - d_j$, and $G_i(t) = G(t \mid \boldsymbol{D}_i)$. On an event observation $R_i = \varepsilon_i$. The conditional event subdensity is $f_\varepsilon(t)G_i(t)$, whereas the conditional survival function of R_j is $S_\varepsilon(t)G_j(t)$. Consequently,

$$A_{ij} := E_D\{\Delta_i \mathbf{1}(Q_{ij} \geq 0)\} = \int f_\varepsilon(t) S_\varepsilon(t) G_i(t) G_j(t)\, \mathrm{d}t = A_{ji}. \tag{C.1}$$

Since $\boldsymbol{w}_{ji} = -\boldsymbol{w}_{ij}$, $N^{-1}\sum_{i\neq j} A_{ij}\boldsymbol{w}_{ij} = 0$. Centering follows from cancellation of the two ordered contributions within each pair.

The finite event-weighted measure $\Pr_D(\Delta_i = 1, Q_{ij} \in \mathrm{d}r)$ has density

$$q_{ij}(r) = \int f_\varepsilon(z - r) G_i(z - r)\, \mathrm{d}F_{R_j \mid \boldsymbol{D}_j}(z) \leq L_\varepsilon \quad \text{a.e.} \tag{C.2}$$

The integral is Stieltjes and permits atoms in R_j. Restricting it to the event submeasure of patient j gives, for $|r| \leq A$,

$$\begin{aligned} q_{ij}(r) &\geq \int f_\varepsilon(z - r) G_i(z - r) f_\varepsilon(z) G_j(z)\, \mathrm{d}z \\ &\geq (2A - |r|) m_\varepsilon^2 g_0^2 \geq c_{\mathrm{loc}}. \end{aligned} \tag{C.3}$$

The intersection of I and $I + r$ has length $2A - |r|$; the location $t_\star$ disappears from the difference interval.

For any real ζ, splitting a shifted smoothed step from an unshifted step yields

$$\int_{\mathbb{R}} |\Phi\{(r + \zeta)/h\} - \mathbf{1}(r \geq 0)|\, \mathrm{d}r \leq |\zeta| + h\sqrt{2/\pi}. \tag{C.4}$$

This follows by adding and subtracting $\mathbf{1}(r + \zeta \geq 0)$, integrating its shift over an interval of length $|\zeta|$, and applying the scalar

smoothing integral from Section 3. The integrated shift bound is linear in $|\zeta|$ with a separate $O(h)$ smoothing term.

Using (C.1)–(C.4), $|\zeta_{ij}| \le 2b$, and Cauchy–Schwarz,

$$|(\boldsymbol{g}_h)_j| \le L_\varepsilon\{2b + h\sqrt{2/\pi}\}\frac{1}{N}\sum_{i\neq l}|w_{il,j}| \\ \le L_\varepsilon\{2b + h\sqrt{2/\pi}\}\sqrt{\widehat{\Gamma}_{jj}} \le 2V_n L_\varepsilon\{2b + h\sqrt{2/\pi}\}.$$

Here $\widehat{\Gamma}_{jj} = 2nV_{n,j}^2/(n-1) \le 4V_n^2$, with $V_{n,j} = \|\widehat{\boldsymbol{W}}_{\cdot j}\|_2/\sqrt{n}$. This proves the mean-score assertion of Lemma 4.1. The same column argument gives $|L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger)| \le 2V_n\|\boldsymbol{v}\|_1$ and justifies conditional contrast integrability without an error moment.

C.2 A column-scale concentration bound for the gradient

Let G_j be coordinate j of $\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger)$. Replacing patient i's outcome changes the $2(n-1)$ ordered terms involving that patient. Each scalar weight $\Delta_l\Phi(\cdot)$ lies in $[0,1]$. A valid bounded-difference constant is therefore

$$c_{ij} = \frac{2}{N}\sum_{l\neq i}|w_{il,j}|.$$

By Cauchy–Schwarz and the pairwise variance identity,

$$\sum_i c_{ij}^2 \le \frac{4(n-1)}{N^2}\sum_{i\neq l} w_{il,j}^2 = \frac{4}{n}\widehat{\Gamma}_{jj} \le \frac{16V_n^2}{n}. \tag{C.5}$$

The two-sided bounded-difference inequality consequently gives

$$\Pr_D\{|G_j - E_D G_j| > t\} \le 2\exp\{-nt^2/(8V_n^2)\}.$$

A union bound over P coordinates proves (4.9). This calculation uses patients, not ordered pairs, as the independent units.

C.3 Pairing and moment contraction for the contrast process

For $i < j$ define the symmetric increment

$$H_{ij}(\boldsymbol{v}) = \tfrac{1}{2}\Big[\Delta_i\{\ell_h(r_{ij}^\dagger + \boldsymbol{w}_{ij}^\top\boldsymbol{v}) - \ell_h(r_{ij}^\dagger)\} \\ + \Delta_j\{\ell_h(-r_{ij}^\dagger - \boldsymbol{w}_{ij}^\top\boldsymbol{v}) - \ell_h(-r_{ij}^\dagger)\}\Big], \tag{C.6}$$

where $r_{ij}^{\dagger} = Q_{ij} + \zeta_{ij}$. Extend $H_{ji} = H_{ij}$. For fixed outcomes, $H_{ij}(\boldsymbol{v}) = g_{ij}(\boldsymbol{w}_{ij}^{\top}\boldsymbol{v})$ with $g_{ij}(0) = 0$ and Lipschitz constant at most one. Moreover,

$$L_{n,h}(\boldsymbol{\vartheta}^{\dagger} + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^{\dagger}) = \binom{n}{2}^{-1} \sum_{i<j} H_{ij}(\boldsymbol{v}).$$

Let $m = \lfloor n/2 \rfloor$ and let π be uniform over permutations of the patient labels, independent of the outcomes. Write

$$U_{\pi}(\boldsymbol{v}) = \frac{1}{m} \sum_{l=1}^{m} H_{\pi(2l-1),\pi(2l)}(\boldsymbol{v}).$$

Every unordered pair is equally likely to appear among the m pairs, so

$$E_{\pi} U_{\pi}(\boldsymbol{v}) = \binom{n}{2}^{-1} \sum_{i<j} H_{ij}(\boldsymbol{v}). \tag{C.7}$$

This identity also holds for odd n, where one label is unused in each permutation. Conditional expectations must be averaged over π as well: $E_{\pi} E_D U_{\pi}(\boldsymbol{v}) = \mathcal{R}_h(\boldsymbol{v})$. This average retains the patient-specific conditional laws.

Set $Z(r) = \sup_{\|\boldsymbol{v}\|_1 \le r} |U(\boldsymbol{v}) - E_D U(\boldsymbol{v})|$, where $U(\boldsymbol{v})$ is the left side of (C.7). For any moment order $a \ge 1$, Jensen and Minkowski, followed by ghost-sample symmetrization of the independent pairs, give

$$\begin{aligned} \|Z(r)\|_{L^a(\boldsymbol{D})} &\le E_{\pi} \left\| \sup_{\|\boldsymbol{v}\|_1 \le r} |U_{\pi}(\boldsymbol{v}) - E_D U_{\pi}(\boldsymbol{v})| \right\|_{L^a(\boldsymbol{D})} \\ &\le \frac{2}{m} E_{\pi} \left\| \sup_{\|\boldsymbol{v}\|_1 \le r} \left| \sum_{l=1}^{m} \sigma_l H_{\pi(2l-1),\pi(2l)}(\boldsymbol{v}) \right| \right\|_{L^a(\boldsymbol{D},\sigma)}, \end{aligned} \tag{C.8}$$

where the σ_l are independent Rademacher signs. The maps g_{ij} may depend on outcomes; after these outcomes are fixed, they are deterministic contractions independent of the signs. The absolute-value moment contraction principle (Ledoux and Talagrand, 1991, Chapter 4), with the safe factor two, and ℓ_1/ℓ_∞ duality yield

$$\|Z(r)\|_{L^a(\boldsymbol{D})} \le \frac{4r}{m} E_{\pi} \left\| \left\| \sum_{l=1}^{m} \sigma_l \boldsymbol{w}_{\pi(2l-1),\pi(2l)} \right\|_{\infty} \right\|_{L^a(\sigma)}. \tag{C.9}$$

Suprema are measurable: the finite-dimensional closed ℓ_1 ball is compact, and the functions are continuous in $\boldsymbol{v}$, so a fixed countable dense subset gives the same supremum.

For every pairing and coordinate j,

$$\sum_{l=1}^{m} w_{\pi(2l-1),\pi(2l),j}^2 \le 2\sum_{i=1}^{n} \widehat{W}_{ij}^2 \le 2nV_n^2. \tag{C.10}$$

Consequently the maximum $T_\pi = \|\sum_l \sigma_l \boldsymbol{w}_{\pi(2l-1),\pi(2l)}\|_\infty$ obeys

$$\Pr_\sigma\{T_\pi > B\sqrt{2\log(2P)} + B\sqrt{2t}\} \le \mathrm{e}^{-t}, \qquad B = \sqrt{2n}\, V_n.$$

Integration of this tail gives

$$\|T_\pi\|_{L^a(\sigma)} \le B\{\sqrt{2\log(2P)} + \sqrt{2a}\} \le 2B\sqrt{\log(2P) + a}. \tag{C.11}$$

For example, the positive part above the first threshold has ath moment at most $(\sqrt{2}B)^a \Gamma(a/2+1)$; the elementary bound $\Gamma(a/2+1)^{1/a} \le \sqrt{a}$ for $a \ge 1$ gives the displayed inequality. Since $m \ge n/3$ for $n \ge 2$, equations (C.9)–(C.11) imply

$$\|Z(r)\|_{L^a(\boldsymbol{D})} \le 24\sqrt{2}\, V_n r\sqrt{\{\log(2P) + a\}/n}. \tag{C.12}$$

Take $a = \log(1/\alpha) > 1$. Markov's inequality at $e\|Z(r)\|_{L^a(\boldsymbol{D})}$ gives failure probability at most $\mathrm{e}^{-a} = \alpha$. Because $24\sqrt{2}e < 96$, this proves (4.10). The proof uses the standard pairing strategy for U-processes (Clémençon et al., 2008), but keeps conditional nonidentical laws and the column norm explicit. Bounding the supremum directly by the largest coordinate would lose this refinement.

C.4 Proof of Lemma 4.2

Differentiation of the conditional contrast is valid for fixed $h > 0$: first and second derivatives of each loss increment are bounded by fixed multiples of $\|\boldsymbol{w}_{ij}\|$ and $\|\boldsymbol{w}_{ij}\|^2/h$. For any real shift z, the scalar second derivative is

$$E_D\{\Delta_i \ell_h''(Q_{ij} + z)\} = \int h^{-1}\phi\{(r+z)/h\} q_{ij}(r)\, \mathrm{d}r. \tag{C.13}$$

If $|z| \le A/2$ and $h \le A/2$, the interval $[-z-h, -z+h]$ is contained in $[-A, A]$. Restricting the integral to that interval and using (C.3)

gives the lower bound $c_{\mathrm{loc}}\{2\Phi(1)-1\} = c_{\mathrm{loc}}c_\phi$. For $t \in [0,1]$ the shift along the parameter segment is

$$z = \zeta_{ij} + t\boldsymbol{w}_{ij}^\top \boldsymbol{v}, \qquad |z| \le 2b + 2M_n\|\boldsymbol{v}\|_1.$$

Thus (4.11) keeps the entire segment in the above interval. Integrating the second derivative along this segment gives

$$\mathcal{R}_h(\boldsymbol{v}) - \boldsymbol{g}_h^\top \boldsymbol{v} \ge \frac{c_{\mathrm{loc}}c_\phi}{2N}\sum_{i\neq j}(\boldsymbol{w}_{ij}^\top \boldsymbol{v})^2.$$

The conditional restricted-design inequality proves the result. The factor c_ϕ comes from retaining one bandwidth on each side of a kernel center; it is not an error-density assumption.

C.5 Localization, finite minimizers, and error bounds

We prove the general version with approximation bound b, which also proves the first part of Corollary 4.1. Put $\mu = \mu(b,h)$ and assume

$$\lambda \ge 4\mu + 384V_n t_\alpha,$$
$$R = 8\sqrt{k}\lambda/\kappa,$$
$$h \le A/2,$$
$$2b + 8M_n\sqrt{k}R \le A/2.$$

Use Lemma 4.1 with the fixed conditional radius $r = 4\sqrt{k}R$. On the intersection of its two events,

$$\|\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger)\|_\infty \le \mu + \sqrt{8}V_n t_\alpha \le \lambda/2.$$

Suppose $\boldsymbol{v}$ satisfies $Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) \le Q(\boldsymbol{\vartheta}^\dagger)$. Convexity of $L_{n,h}$ and the support of $\boldsymbol{\beta}_0$ imply

$$\begin{aligned} 0 &\ge Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - Q(\boldsymbol{\vartheta}^\dagger) \\ &\ge -\tfrac{\lambda}{2}\|\boldsymbol{v}\|_1 + \lambda\{\|(\boldsymbol{v}_\beta)_{S^c}\|_1 - \|(\boldsymbol{v}_\beta)_S\|_1\}. \end{aligned}$$

Rearrangement proves (4.16), and therefore $\boldsymbol{v} \in \mathcal{C}(S)$ and $\|\boldsymbol{v}\|_1 \le 4\sqrt{k}\|\boldsymbol{v}\|_2$.

Now assume additionally that $\|\boldsymbol{v}\|_2 = R$. The local curvature condition holds, and the risk contrast, empirical contrast bound, and penalty inequality give

$$\begin{aligned} Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - Q(\boldsymbol{\vartheta}^\dagger) &\geq \tfrac{\kappa}{2}R^2 - \mu\|\boldsymbol{v}\|_1 - 96V_n r t_\alpha - \lambda\|(\boldsymbol{v}_\beta)_S\|_1 \\ &\geq \tfrac{\kappa}{2}R^2 - \{4\mu + 384V_n t_\alpha + \lambda\}\sqrt{k}R \\ &\geq \tfrac{\kappa}{2}R^2 - 2\lambda\sqrt{k}R = \tfrac{\kappa}{4}R^2 > 0. \end{aligned} \tag{C.14}$$

This contradicts the assumed sublevel inequality. Thus no point in the sublevel set through $\boldsymbol{\vartheta}^\dagger$ lies on the sphere of radius R.

If a point of that sublevel set were outside the ball, its line segment to $\boldsymbol{\vartheta}^\dagger$, by convexity of Q, would give a point on the excluded sphere. Hence the whole sublevel set is inside the ball. It is nonempty and closed, so it is compact. The continuous objective attains a minimum there, and every point outside the sublevel set has a strictly larger value than $Q(\boldsymbol{\vartheta}^\dagger)$. This minimum is therefore global. All global minimizers lie in that compact set and satisfy

$$\|\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 \leq R, \qquad \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 \leq 4\sqrt{k}R.$$

The set of minimizers is closed and compact as well. Its nonemptiness is proved rather than assumed. Substituting R gives (4.15); the event has conditional probability at least $1 - 2\alpha$. Setting $b = 0$ proves Theorem 4.1. Using the representation (4.28) proves the conditional extension.

D Primitive factor-design guarantees

Constants $C, c > 0$ in this appendix can change between displays and depend only on the fixed constants of Assumption 4.2 and on d, K. Dependence on the outcome constants is added only in the final estimation step. The probability parameter is δ, and $L_\delta, \tau_\delta, a_\delta, r_{F,\delta}$ are defined in (4.19).

D.1 One joint covariate event

Let $\boldsymbol{D}$ be the matrix with rows $\boldsymbol{D}_i^\top$ and $\boldsymbol{D}_c = \boldsymbol{J}\boldsymbol{D}$. We first construct an event of probability at least $1-\delta$ on which

$$\|\boldsymbol{D}_c^\top \boldsymbol{D}_c/n - \boldsymbol{\Sigma}_D\|_{\max} \le C\tau_\delta, \qquad \max_{i,j} |D_{ij}| \le C\sqrt{L_\delta}, \tag{D.1}$$

$$\|\boldsymbol{U}_c\|_{\mathrm{op}} \le C(\sqrt{n}+\sqrt{p}), \qquad \max_i \|\boldsymbol{u}_{c,i}\|_2 \le C\sqrt{p}, \tag{D.2}$$

$$\max_i \|\boldsymbol{B}^\top \boldsymbol{u}_{c,i}\|_2 \le C\sqrt{pL_\delta}, \qquad c\boldsymbol{I}_K \preceq \boldsymbol{F}_c^\top \boldsymbol{F}_c/n \preceq C\boldsymbol{I}_K. \tag{D.3}$$

Here $\boldsymbol{u}_{c,i}$ is row i of $\boldsymbol{U}_c$ written as a column. The prerequisite is $n \wedge p \ge CL_\delta$. The following concentration bounds construct this joint design event.

For every coordinate j, sub-Gaussian tails and a union bound over nP variables give the maximum in (D.1). The sample means obey $\max_j |\bar{\boldsymbol{D}}_j| \le C\sqrt{L_\delta/n}$. Products of two uniformly sub-Gaussian coordinates are uniformly sub-exponential; this does not require independence between the coordinates. Bernstein's inequality and a union bound over P^2 products give

$$\max_{j,l} |n^{-1}\sum_i D_{ij}D_{il} - \boldsymbol{\Sigma}_{D,jl}| \le C\{\sqrt{L_\delta/n} + L_\delta/n\}.$$

Subtracting $\bar{\boldsymbol{D}}\bar{\boldsymbol{D}}^\top$ proves the covariance part of (D.1). Since K is fixed and $\boldsymbol{\Sigma}_f = \boldsymbol{I}_K$, its factor block gives the final spectral statement of (D.3) when the size constant is sufficiently large. It also gives

$$\max_j \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n} \le C, \qquad \max_j \|\boldsymbol{F}_c^\top \boldsymbol{U}_{c,\cdot j}/n\|_2 \le C\tau_\delta, \tag{D.4}$$

because $E(\boldsymbol{f}_i \boldsymbol{u}_i^\top) = 0$.

For completeness, a bilinear net argument controls the noise operator norm even though coordinates within each $\boldsymbol{u}_i$ may be dependent. For fixed unit vectors $\boldsymbol{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}^p$, $\boldsymbol{a}^\top \boldsymbol{U} b = \sum_i a_i b^\top \boldsymbol{u}_i$ is centered sub-Gaussian with a fixed norm, by independence across patients and the joint sub-Gaussian condition. Apply its tail bound to 1/4-nets of the two spheres, with at most 9^n and 9^p points. The net approximation and a union bound yield

$$\|\boldsymbol{U}\|_{\mathrm{op}} \le C\{\sqrt{n} + \sqrt{p} + \sqrt{\log(1/\delta)}\}.$$

Centering is a contraction, and $n \wedge p \ge C L_\delta$ absorbs the last term. A 1/2-net for a single row, followed by a union bound over patients, gives $\max_i \|\boldsymbol{u}_i\|_2 \le C(\sqrt{p} + \sqrt{L_\delta})$. Since $\|\bar{\boldsymbol{u}}\|_2 \le \max_i \|\boldsymbol{u}_i\|_2$, centering costs at most a factor two, proving (D.2). These are standard net and concentration tools for jointly sub-Gaussian vectors (Vershynin, 2012).

Each coordinate of $\boldsymbol{B}^\top \boldsymbol{u}_i$ has sub-Gaussian norm at most $C\sqrt{p}$, because $\|\boldsymbol{B}_{\cdot l}\|_2 \le C\sqrt{p}$. A union bound over the fixed K projections and n patients, with the same centering argument, proves the first part of (D.3). Assigning the constituent events fixed fractions of δ and increasing C makes their intersection have probability at least $1 - \delta$. All statements below are deterministic on this intersection. In particular, $\boldsymbol{H}$ may be a function of the same data; the bounds are uniform over the relevant vectors and do not require $\boldsymbol{H}$ to be independent.

D.2 Spectral separation and an explicit alignment

The signal matrix $\boldsymbol{F}_c \boldsymbol{B}^\top$ has rank K and nonzero singular values between $c\sqrt{np}$ and $C\sqrt{np}$. The perturbation $\boldsymbol{U}_c$ has norm at most $C(\sqrt{n} + \sqrt{p})$. The singular-value perturbation inequality therefore implies

$$c\sqrt{np} \le s_K(\boldsymbol{X}_c) \le s_1(\boldsymbol{X}_c) \le C\sqrt{np}, \qquad s_{K+1}(\boldsymbol{X}_c) \le \|\boldsymbol{U}_c\|_{\mathrm{op}},$$

where the last singular value is read as zero if the index exceeds the dimension. With sufficiently large $n \wedge p$, the latter is strictly below $s_K(\boldsymbol{X}_c)$. Thus $cp\boldsymbol{I}_K \preceq \widehat{\boldsymbol{\Lambda}} \preceq Cp\boldsymbol{I}_K$ and there is a gap at the selected rank.

The PCA identities give, without any probabilistic approximation,

$$\widehat{\boldsymbol{F}} = \boldsymbol{X}_c\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} = \boldsymbol{F}_c\boldsymbol{B}^\top\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} + \boldsymbol{U}_c\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} = \boldsymbol{F}_c\boldsymbol{H}^\top + \boldsymbol{E}_F,$$

with $\boldsymbol{H}, \boldsymbol{E}_F$ from (4.20). Since $\widehat{\boldsymbol{B}}^\top\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{\Lambda}}$,

$$\|\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}\|_{\mathrm{op}} = \lambda_{\min}(\widehat{\boldsymbol{\Lambda}})^{-1/2} \le C/\sqrt{p}.$$

It follows that

$$\frac{\|\boldsymbol{E}_F\|_{\mathrm{op}}}{\sqrt{n}} \le C\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{p}}\right). \tag{D.5}$$

This derivation does not first assume that $\boldsymbol{H}$ is invertible. Instead, $\widehat{\boldsymbol{F}}$ has all singular values $\sqrt{n}$, and (D.5) makes $\boldsymbol{F}_c\boldsymbol{H}^\top = \widehat{\boldsymbol{F}} - \boldsymbol{E}_F$ have all its nonzero singular values bounded above and below by fixed multiples of $\sqrt{n}$. Since $\boldsymbol{F}_c$ has the same scale, $\boldsymbol{H}$ must be invertible, with $\|\boldsymbol{H}\|_{\mathrm{op}} + \|\boldsymbol{H}^{-1}\|_{\mathrm{op}} \le C$. Sample alignment is not assumed orthogonal, consistently with the normalization distinction discussed by Bai and Ng (2013).

D.3 Maximum-row factor error and maximum-column residual error

Set $\boldsymbol{A}_F = \boldsymbol{F}_c^\top\widehat{\boldsymbol{F}}/n$, so $\|\boldsymbol{A}_F\|_{\mathrm{op}} \le C$ and

$$\widehat{\boldsymbol{B}} = \boldsymbol{B}\boldsymbol{A}_F + \boldsymbol{U}_c^\top\widehat{\boldsymbol{F}}/n.$$

For a row error written as a column,

$$e_{F,i} = \widehat{\boldsymbol{\Lambda}}^{-1}\{\boldsymbol{A}_F^\top\boldsymbol{B}^\top\boldsymbol{u}_{c,i} + \widehat{\boldsymbol{F}}^\top\boldsymbol{U}_c\boldsymbol{u}_{c,i}/n\}. \tag{D.6}$$

The first term is at most $C\sqrt{L_\delta/p}$. For the second, apply only deterministic norm inequalities:

$$\begin{aligned}\frac{C}{p}\frac{\|\widehat{\boldsymbol{F}}\|_{\mathrm{op}}}{n}\|\boldsymbol{U}_c\|_{\mathrm{op}}\max_i\|\boldsymbol{u}_{c,i}\|_2 &\le \frac{C}{p}\frac{1}{\sqrt{n}}(\sqrt{n}+\sqrt{p})\sqrt{p}\\ &\le C(p^{-1/2}+n^{-1/2}).\end{aligned}$$

Combining the two proves the maximum-row bound in (4.21). The calculation is valid with same-sample factor estimation: no independence between $\widehat{\boldsymbol{F}}$ and $\boldsymbol{U}_c$ was used.

Let $\widehat{P}_F = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{F}}^\top/n$ and let $\boldsymbol{b}_j$ be column j of $\boldsymbol{B}^\top$. The residual difference satisfies

$$\widehat{\boldsymbol{U}}_{\cdot j} - \boldsymbol{U}_{c,\cdot j} = (I - \widehat{P}_F)\boldsymbol{F}_c\boldsymbol{b}_j - \widehat{P}_F\boldsymbol{U}_{c,\cdot j}. \tag{D.7}$$

The first term equals $-(I - \widehat{P}_F)\boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{b}_j$ and, divided by $\sqrt{n}$, is at most $C(n^{-1/2} + p^{-1/2})$, because $\|\boldsymbol{b}_j\|_2 \le L_B$. For the second term,

$$\begin{aligned}\frac{\|\widehat{P}_F\boldsymbol{U}_{c,\cdot j}\|_2}{\sqrt{n}} &= \left\|\frac{\widehat{\boldsymbol{F}}^\top\boldsymbol{U}_{c,\cdot j}}{n}\right\|_2 \\ &\le \|\boldsymbol{H}\|_{\mathrm{op}}\left\|\frac{\boldsymbol{F}_c^\top\boldsymbol{U}_{c,\cdot j}}{n}\right\|_2 + \frac{\|\boldsymbol{E}_F\|_{\mathrm{op}}}{\sqrt{n}}\frac{\|\boldsymbol{U}_{c,\cdot j}\|_2}{\sqrt{n}} \\ &\le C\{\tau_\delta + p^{-1/2}\}.\end{aligned}$$

Here $n^{-1/2} \le \tau_\delta$. This proves the maximum-column residual bound in (4.22). Factor-error operator control and residual column control imply

$$\begin{gathered}\max_{j\le P}\frac{\|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2}{\sqrt{n}} \le Ca_\delta, \\ \boldsymbol{W} = (\boldsymbol{Z}_c, \boldsymbol{F}_c\boldsymbol{H}^\top, \boldsymbol{U}_c).\end{gathered} \tag{D.8}$$

The clinical block has zero approximation error.

To control the scales in the conditional theorem, first note

$$\max_j \|\boldsymbol{X}_{c,\cdot j}\|_2/\sqrt{n} \le \max_j\{\|\boldsymbol{F}_c\|_{\mathrm{op}}\|\boldsymbol{b}_j\|_2/\sqrt{n} + \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n}\} \le C.$$

Since $\widehat{\boldsymbol{U}} = (I - \widehat{P}_F)\boldsymbol{X}_c$, its column norms are no larger. The factor columns have norm $\sqrt{n}$, and clinical columns are bounded using (D.1). Thus $V_n \le C$. Also

$$\max_j \|\widehat{\boldsymbol{B}}_{j\cdot}\|_2 \le \max_j\{\|\boldsymbol{b}_j\|_2\|\boldsymbol{A}_F\|_{\mathrm{op}} + \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n}\} \le C.$$

Using $\widehat{\boldsymbol{F}} = \boldsymbol{F}_c\boldsymbol{H}^\top + \boldsymbol{E}_F$ and the maximum-row bound gives $\max_i \|\hat{\boldsymbol{f}}_i\|_2 \le C\sqrt{L_\delta}$. Together with $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ and the row-loading bound, this yields $M_n \le C\sqrt{L_\delta}$. If the raw $\boldsymbol{D}_i$ coordinates are uniformly bounded, the centered $\boldsymbol{F}_c, \boldsymbol{U}_c, \boldsymbol{Z}_c$ coordinates are bounded, while $r_{F,\delta}$ is bounded under $n \wedge p \ge CL_\delta$. The same argument then gives $M_n \le C$. Coordinate boundedness is an additional assumption, not a consequence of observing a finite data matrix.

D.4 Ideal geometry and proof of the transfer lemma

Write $\boldsymbol{W} = \boldsymbol{D}_c\boldsymbol{T}_H$ with $\boldsymbol{T}_H = \mathrm{diag}(\boldsymbol{I}_d, \boldsymbol{H}^\top, \boldsymbol{I}_p)$. On the joint event, $\|\boldsymbol{T}_H\boldsymbol{v}\|_2 \geq c\|\boldsymbol{v}\|_2$ and $\|\boldsymbol{T}_H\boldsymbol{v}\|_1 \leq C\|\boldsymbol{v}\|_1$ because K is fixed and $\boldsymbol{H}, \boldsymbol{H}^{-1}$ are controlled. These inequalities hold for the data-dependent $\boldsymbol{T}_H$ pointwise. The covariance event gives

$$\begin{aligned}\|\boldsymbol{W}\boldsymbol{v}\|_2^2/n &\geq (\boldsymbol{T}_H\boldsymbol{v})^\top \boldsymbol{\Sigma}_D(\boldsymbol{T}_H\boldsymbol{v}) - C\tau_\delta\|\boldsymbol{T}_H\boldsymbol{v}\|_1^2 \\ &\geq (c_0 - C_0 k\tau_\delta)\|\boldsymbol{v}\|_2^2, \qquad \boldsymbol{v} \in \mathcal{C}(S). \end{aligned} \tag{D.9}$$

For $k\tau_\delta$ sufficiently small this is bounded below by a fixed multiple of $\|\boldsymbol{v}\|_2^2$. Centering yields $\|\boldsymbol{W}\boldsymbol{v}\|_{\mathrm{pair}}^2 = 2\|\boldsymbol{W}\boldsymbol{v}\|_2^2/(n-1)$, so the ideal pairwise restricted bound holds with some fixed $\kappa_0 > 0$.

For any vector $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_{\mathrm{pair}}^2 = \frac{2}{n-1}\|\boldsymbol{J}\boldsymbol{x}\|_2^2. \tag{D.10}$$

Under the premise of Lemma 4.3,

$$\|(\widehat{\boldsymbol{W}} - \boldsymbol{W})\boldsymbol{v}\|_2/\sqrt{n} \leq \sum_j |v_j|\|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2/\sqrt{n} \leq a\|\boldsymbol{v}\|_1.$$

Equation (D.10) and $n \geq 2$ give

$$\|(\widehat{\boldsymbol{W}} - \boldsymbol{W})\boldsymbol{v}\|_{\mathrm{pair}} \leq 2a\|\boldsymbol{v}\|_1 \leq 8a\sqrt{k}\|\boldsymbol{v}\|_2 \quad (\boldsymbol{v} \in \mathcal{C}(S)).$$

The triangle inequality proves (4.24). Combine it with (D.8) and (D.9). If $\sqrt{k}a_\delta$ is sufficiently small, the resulting lower bound is $\kappa_X \geq c_X > 0$. This completes Proposition 4.1. The max-norm covariance argument yields the sufficient requirement on $k\tau_\delta$.

D.5 Integrating the conditional result

On $\mathcal{E}_D$ in the ordinary model, $b_n = 0$, $V_n \leq C$, $M_n \leq C\sqrt{L_\delta}$, and the curvature constant has a fixed positive lower bound. Apply Theorem 4.1 with $\alpha = \delta$. Since $t_\alpha \leq t_\delta$ and $\mu(0,h) \leq Ch$, sufficiently large C_* in (4.25) makes the conditional penalty inequality hold. The condition $k\sqrt{L_\delta}\lambda \leq c_*$ ensures

$$8M_n\sqrt{k}R = 64M_nk\lambda/\kappa \leq A/2$$

for sufficiently small c_*. In the bounded-coordinate case, $k\lambda \le c_*$ suffices. The conditional bounds become (4.26).

The total failure probability is at most

$$\Pr(\mathcal{E}_D^c) + E\{\mathbf{1}_{\mathcal{E}_D} \Pr(\text{outcome failure} \mid D_n)\} \le \delta + 2\delta = 3\delta.$$

No independence between the design event and the outcome event is needed. The theorem therefore controls the feasible estimator, not only a latent-design minimizer.

For Corollary 4.1, $b_n \le \max_i \|e_{F,i}\|_2 \|\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\|_2 \le C\Xi r_{F,\delta}$ on $\mathcal{E}_D$. The penalty level in (4.30) dominates $4\mu(b_n, h) + 384V_n t_\alpha$. Its locality inequality ensures $2b_n + 64M_n k\lambda/\kappa \le A/2$. The conditional extension proved in Appendix C.5, followed by the same probability integration, gives (4.31).

Finally, with $p \ge n$ and $h \le Ct_\delta$, we have $\tau_\delta \lesssim t_\delta$ when L_δ/n is small, and $a_\delta \lesssim t_\delta$. The jointly sub-Gaussian condition $k^2 L_\delta^2/n \to 0$ implies both $k\tau_\delta \to 0$ and $\sqrt{k}a_\delta \to 0$, and ensures locality. With bounded coordinates, $k^2 L_\delta/n \to 0$ suffices. For the compatible example in the text, independent Rademacher coordinates have identity covariance and a uniformly bounded joint sub-Gaussian norm by the product moment generating function. The Cauchy density is globally bounded and is at least $1/(2\pi)$ on $[-1, 1]$; $C_i^* = 2$ gives $G_i = 1$ there. Positive censoring probability is $\Pr(\varepsilon_i > 2) > 0$. Taking $p = n^2$, $k = \lfloor n^{1/4} \rfloor + d + 1$, and $\delta_n = n^{-2}$ makes $k^2 L_{\delta_n}/n \to 0$. This verifies compatibility without a finite error mean or variance.

E Computational identities and convergence proofs

E.1 Pairwise derivatives, Laplacian bound, and implicit products

Let $\boldsymbol{w}_{ij} = \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j$. Since $\partial r_{ij}/\partial \boldsymbol{\vartheta} = \boldsymbol{w}_{ij}$, differentiation of a finite sum gives

$$\nabla f_h(\boldsymbol{\vartheta}) = \frac{1}{N}\sum_{i\neq j} \Delta_i \Phi(r_{ij}/h)\boldsymbol{w}_{ij},$$

$$\nabla^2 f_h(\boldsymbol{\vartheta}) = \frac{1}{Nh}\sum_{i\neq j} \Delta_i \phi(r_{ij}/h)\boldsymbol{w}_{ij}\boldsymbol{w}_{ij}^\top.$$

Collecting the positive coefficient of row i and the negative coefficient of column i gives (5.2). Summing c_i over i gives zero. Blocking the row sum does not change any term. Only rows with $\Delta_i = 1$ need to be evaluated, but each such row still includes all other patients.

Combining the (i, j) and (j, i) outer products yields

$$\sum_{i\neq j} \Delta_i \boldsymbol{w}_{ij}\boldsymbol{w}_{ij}^\top = \sum_{i<j} (\Delta_i + \Delta_j)\boldsymbol{w}_{ij}\boldsymbol{w}_{ij}^\top = \widehat{\boldsymbol{W}}^\top \mathscr{L}_{\Delta} \widehat{\boldsymbol{W}}.$$

The diagonal of this Laplacian is $(n-2)\Delta_i + m$, and its off-diagonal entry is $-(\Delta_i + \Delta_j)$, proving (5.4). The bound $\phi \leq 1/\sqrt{2\pi}$ then proves (5.5), with the Lipschitz property obtained by integrating the Hessian along the segment between two finite points. Centering removes both terms involving $\widehat{\boldsymbol{W}}^\top \mathbf{1}_n$. The operator-norm bound follows from $0 \preceq \operatorname{diag}(\boldsymbol{\Delta}) \preceq \boldsymbol{I}_n$.

For the positive line-search certificate, Taylor's integral remainder is

$$\begin{aligned} &f_h(\boldsymbol{y}) - f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) \\ &\quad = \frac{1}{Nh}\sum_{i\neq j} \Delta_i d_{ij}^2 \int_0^1 (1-s)\phi\{(r_{ij}(\boldsymbol{x}) + s d_{ij})/h\}\,\mathrm{d}s. \end{aligned}$$

The normal density decreases as its argument's absolute value increases. Its maximum on the segment is therefore $\phi(m_{ij}/h)$. Bounding the integral by half that maximum proves the first inequality in (5.8); bounding it by $\phi(0)/2$ and using the Laplacian proves the second. Thus this alternative test also accepts every $t \leq \omega/\mathscr{B}_h$, which is the only threshold property needed in the convergence proof.

For completeness, Hessian-vector products can be computed without forming the Hessian. Let $t_i = \widehat{\boldsymbol{W}}_i^\top \boldsymbol{v}$ and, for $i \neq j$, define

$$R_{ij} = \frac{\Delta_i}{h}\phi(r_{ij}/h)(t_i - t_j), \qquad R_{ii} = 0, \qquad b_i = \sum_j R_{ij} - \sum_j R_{ji}.$$

Then $\nabla^2 f_h(\boldsymbol{\vartheta})\boldsymbol{v} = \widehat{\boldsymbol{W}}^\top b/N$. This formula is useful for testing, although Algorithm 1 does not require a Hessian-vector product. Substituting $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ proves both identities in (5.3).

E.2 Proximal optimality and the three-point inequality

For $\boldsymbol{y} = T_t(\boldsymbol{x})$, strict convexity of the proximal quadratic ensures a unique candidate and gives

$$(\boldsymbol{x} - \boldsymbol{y})/t - \nabla f_h(\boldsymbol{x}) \in \partial g_\lambda(\boldsymbol{y}). \tag{E.1}$$

Separability produces the identity update on the unpenalized block and soft thresholding on the penalized block in (5.6). For any finite $\boldsymbol{u}$, the subgradient inequality, convexity of f_h, and the accepted line search give

$$\begin{aligned} Q(\boldsymbol{y}) - Q(\boldsymbol{u}) &\leq \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{u}) + \frac{\omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2 + g_\lambda(\boldsymbol{y}) - g_\lambda(\boldsymbol{u}) \\ &\leq \frac{(\boldsymbol{x} - \boldsymbol{y})^\top(\boldsymbol{y} - \boldsymbol{u})}{t} + \frac{\omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \end{aligned}$$

Using $2(\boldsymbol{x} - \boldsymbol{y})^\top(\boldsymbol{y} - \boldsymbol{u}) = \|\boldsymbol{x} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{x}\|_2^2$ yields

$$Q(\boldsymbol{y}) - Q(\boldsymbol{u}) \leq \frac{\|\boldsymbol{x} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{u}\|_2^2}{2t} - \frac{1 - \omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \tag{E.2}$$

In particular, taking $\boldsymbol{u} = \boldsymbol{x}$ proves (5.10), including its factor $2 - \omega$. This uses exact minimization of the proximal surrogate, not merely comparison of that surrogate at $\boldsymbol{x}$ and $\boldsymbol{y}$.

E.3 Proof of Theorem 5.1

The global smoothness bound gives

$$f_h(\boldsymbol{y}) \leq f_h(\boldsymbol{x}) + \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mathscr{B}_h}{2}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2.$$

Hence the line search must accept whenever $t \le \omega/\mathscr{B}_h$ if $\mathscr{B}_h > 0$. Starting from $t_{\max}$ and multiplying by ζ therefore takes finitely many trials. If a reduction occurs before acceptance, the previously rejected step exceeds $\omega/\mathscr{B}_h$, proving the lower bound (5.9); acceptance at the first trial gives $t_r = t_{\max}$. If $\mathscr{B}_h = 0$, the smooth objective is affine and the first trial is accepted. For this particular nonnegative pairwise loss it is constant, but affineness is already sufficient for acceptance.

Fix $\boldsymbol{x}^\star \in \mathcal{S}$. Applying (E.2) with $\boldsymbol{u} = \boldsymbol{x}^\star$ gives

$$2t_r\{Q(\boldsymbol{x}^{(r+1)}) - Q(\boldsymbol{x}^\star)\} \le \|\boldsymbol{x}^{(r)} - \boldsymbol{x}^\star\|_2^2 - \|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^\star\|_2^2 - (1-\omega)\|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2. \tag{E.3}$$

The left side is nonnegative, so distances to every minimizer are nonincreasing and the sequence is bounded. Summing this inequality for $r = 0, \dots, R-1$ and using monotonicity of Q yields

$$2Rt_{\min}\{Q(\boldsymbol{x}^{(R)}) - Q(\boldsymbol{x}^\star)\} \le \|\boldsymbol{x}^{(0)} - \boldsymbol{x}^\star\|_2^2,$$

which proves (5.11). Summing the descent inequality and using $t_r \le t_{\max}$ and $Q \ge 0$ gives

$$\sum_{r=0}^{\infty} \|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2 < \infty.$$

Since $t_r \ge t_{\min} > 0$, $G_{t_r}(\boldsymbol{x}^{(r)}) \to 0$. By (E.1), the vectors

$$s_{r+1} = G_{t_r}(\boldsymbol{x}^{(r)}) + \nabla f_h(\boldsymbol{x}^{(r+1)}) - \nabla f_h(\boldsymbol{x}^{(r)}) \in \partial Q(\boldsymbol{x}^{(r+1)})$$

converge to zero, using global Lipschitz continuity. A convergent subsequence exists by boundedness. If its limit is $\bar{\boldsymbol{x}}$, continuity of Q and the subgradient inequality imply, for every finite $\boldsymbol{u}$,

$$Q(\boldsymbol{u}) \ge Q(\boldsymbol{x}^{(r_j)}) + s_{r_j}^\top(\boldsymbol{u} - \boldsymbol{x}^{(r_j)}) \longrightarrow Q(\bar{\boldsymbol{x}}).$$

Thus $\bar{\boldsymbol{x}} \in \mathcal{S}$. Distances to this particular minimizer are nonincreasing and have a subsequence tending to zero, so the whole sequence converges to $\bar{\boldsymbol{x}}$.

E.4 Optimality residuals and proof of Corollary 5.1

For a nonzero β_j, the jth subgradient coordinate is the singleton $g_{\beta,j} + \lambda\,\mathrm{sign}(\beta_j)$; for $\beta_j = 0$, it is the interval $[g_{\beta,j} - \lambda, g_{\beta,j} + \lambda]$.

The nuisance subgradient is always $\boldsymbol{g}_a$. Minimizing the absolute value coordinatewise proves (5.12), and the minimum is attained. Exact zero of this residual is equivalent to convex optimality. The fixed-point equivalence for G_t follows from (E.1). Adding $\nabla f_h(\boldsymbol{y})$ there proves membership in (5.14); its norm bound follows by the triangle inequality and $\|\boldsymbol{x}-\boldsymbol{y}\|_2 = t\|G_t(\boldsymbol{x})\|_2$.

We next make explicit the Section 4 event used for the finite-tolerance transfer. Put $t_\alpha = \{\log(2P/\alpha)/n\}^{1/2}$ and $\mu = 2V_n L_\varepsilon h\sqrt{2/\pi}$. The tuning condition is $\lambda \geq 4\mu + 384 V_n t_\alpha$. On the score event of that section, the slightly sharper bound

$$\|\nabla f_h(\boldsymbol{\vartheta}^\dagger)\|_\infty \leq \mu + \sqrt{8}\, V_n t_\alpha \leq \lambda/4 \tag{E.4}$$

therefore holds; the earlier proof only needed $\lambda/2$. On its contrast event and under its local curvature assumptions, every $\boldsymbol{u} \in \mathcal{C}(S)$ with $\|\boldsymbol{u}\|_2 = R = 8\sqrt{k}\lambda/\kappa$ satisfies

$$\begin{aligned} Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{u}) - Q(\boldsymbol{\vartheta}^\dagger) &\geq \frac{\kappa}{2}R^2 - (4\mu + 384 V_n t_\alpha + \lambda)\sqrt{k}R \\ &\geq \frac{\kappa}{2}R^2 - 2\lambda\sqrt{k}R = 2\lambda\sqrt{k}R. \end{aligned} \tag{E.5}$$

This inequality holds on the cone sphere itself; a sublevel inequality is not required to derive the displayed lower bound. The empirical contrast is controlled uniformly on the radius-$4\sqrt{k}R$ ℓ_1 ball. No stronger statistical event has been introduced.

Choose $\boldsymbol{s} \in \partial Q(\widetilde{\boldsymbol{\vartheta}})$ with $\|s\|_\infty \leq \varepsilon \leq \lambda/4$, and let $\boldsymbol{v} = \widetilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger$. Convexity at $\widetilde{\boldsymbol{\vartheta}}$ gives the tilted inequality

$$Q(\widetilde{\boldsymbol{\vartheta}}) - Q(\boldsymbol{\vartheta}^\dagger) \leq \boldsymbol{s}^\top \boldsymbol{v} \leq \varepsilon\|\boldsymbol{v}\|_1. \tag{E.6}$$

Convexity of f_h at $\boldsymbol{\vartheta}^\dagger$ and the support of $\boldsymbol{\beta}_0$ also give

$$Q(\widetilde{\boldsymbol{\vartheta}}) - Q(\boldsymbol{\vartheta}^\dagger) \geq -\frac{\lambda}{4}\|\boldsymbol{v}\|_1 + \lambda\{\|(\boldsymbol{v}_\beta)_{S^c}\|_1 - \|(\boldsymbol{v}_\beta)_S\|_1\}.$$

Together with $\varepsilon \leq \lambda/4$, this implies

$$\|(\boldsymbol{v}_\beta)_{S^c}\|_1 \leq 3\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1, \quad \boldsymbol{v} \in \mathcal{C}(S), \quad \|\boldsymbol{v}\|_1 \leq 4\sqrt{k}\|\boldsymbol{v}\|_2.$$

If $\|\boldsymbol{v}\|_2 \geq R$, let $\boldsymbol{u} = R\boldsymbol{v}/\|\boldsymbol{v}\|_2$. Convexity on the segment and (E.6) yield

$$Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{u}) - Q(\boldsymbol{\vartheta}^\dagger) \leq \varepsilon\|\boldsymbol{u}\|_1 \leq \lambda\sqrt{k}R,$$

contradicting (E.5). Hence $\|\boldsymbol{v}\|_2 \le R$ and $\|\boldsymbol{v}_\beta\|_1 \le 4\sqrt{k}R$. This is a pointwise deterministic implication on the original event, so it also applies to an outcome-dependent finite iterate satisfying the residual condition. The choices of K, h, λ must still satisfy the original fixed-tuning conditions; outcome-dependent iteration stopping does not justify outcome-dependent tuning.

E.5 Proof of Proposition 5.1 and the linear test

Let $\mathcal{P} = \{(i,j) : \Delta_i = 1,\ j \ne i\}$. Suppose a strict escape vector $\boldsymbol{v}$ exists. For every finite $(\boldsymbol{a}, \boldsymbol{\beta})$, the derivative of $Q(\boldsymbol{a} + t\boldsymbol{v}, \boldsymbol{\beta})$ with respect to t is

$$\frac{1}{N} \sum_{(i,j)\in\mathcal{P}} \Phi\{r_{ij}(\boldsymbol{a} + t\boldsymbol{v}, \boldsymbol{\beta})/h\}\, \boldsymbol{a}_{ij}^\top \boldsymbol{v} < 0.$$

Every summand is nonpositive and at least one is strictly negative, because the normal distribution function lies strictly between zero and one at a finite argument. The penalty does not change. Therefore no finite point is a minimizer.

Conversely, assume no strict escape vector exists. If $\mathcal{N}_A^\perp$ has positive dimension, define

$$\chi = \min_{\substack{v\in\mathcal{N}_A^\perp \\ \|v\|_2=1}} \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} (\boldsymbol{a}_{ij}^\top v)_+.$$

The continuous function in this display is strictly positive on its compact domain. Indeed, a zero value would give $\boldsymbol{a}_{ij}^\top v \le 0$ for all pairs. The absence of strict escape would then force all products to be zero, so $v \in \mathcal{N}_A$, a contradiction. Thus $\chi > 0$.

Project $\boldsymbol{a}$ onto $\mathcal{N}_A^\perp$, writing $\boldsymbol{a}_\perp$ for this component. All pair predictions, and hence Q, are unchanged. On a nonempty sublevel set $Q \le c$, nonnegativity of ℓ_h implies $\|\boldsymbol{\beta}\|_1 \le c/\lambda$. Since $\ell_h(x+z) \ge (x+z)_+ \ge x_+ - |z|$,

$$f_h(\boldsymbol{a}, \boldsymbol{\beta}) \ge \chi \|\boldsymbol{a}_\perp\|_2 - C_Y - C_U \|\boldsymbol{\beta}\|_1,$$

where

$$C_Y = \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} |Y_j - Y_i|, \qquad C_U = \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} \|\widehat{u}_i - \widehat{u}_j\|_\infty$$

are finite. It follows that $\|\boldsymbol{a}_{\perp}\|_2 \le (c+C_Y+C_U c/\lambda)/\chi$. The projected sublevel set is closed and bounded, and thus compact. If $\mathcal{N}_A^{\perp}$ has dimension zero, $\boldsymbol{a}_{\perp} = 0$ and only the already bounded $\boldsymbol{\beta}$ block remains. Minimizing on the sublevel set through the origin gives a global minimizer in both cases. This covers $q = 0$ as well as an empty event-pair set. Flat directions may produce an unbounded set of equivalent minimizers in the original coordinates; they do not obstruct existence.

For $m > 0$, comparisons between two event patients are present in both directions. Condition (5.17) therefore forces their nuisance scores along v to be equal. Choose an event anchor i_0. All remaining conditions say that censored scores are at least the event score, with at least one strict inequality. The sum of these nonnegative margins is strictly positive and can be scaled to at least one. This proves equivalence with (5.18). In the flat case all sample nuisance scores agree; since $\widehat{\boldsymbol{A}}$ is centered, that common score is zero. Thus, for $m > 0$, $\mathcal{N}_A = \ker(\widehat{\boldsymbol{A}})$.

Finite attainment and approximate stationarity are separated by the following example. Take $n = 2$, nuisance rows $-1, 1$, $\Delta = (1, 0)^{\top}$, and no effective penalized predictor. The objective is $Q(a) = \ell_h(y - 2a)/2$. It has no finite minimizer, but $|Q'(a)| = \Phi\{(y - 2a)/h\} \to 0$ as $a \to \infty$. Any positive gradient tolerance will eventually be met along this escaping sequence. This is why a small residual is not, by itself, evidence of finite attainment outside the statistical event of Section 4.

E.6 Null-fit threshold and stable tail formula

If $\boldsymbol{a}^{(0)}$ is a finite nuisance-only minimizer, differentiability and convexity imply $\nabla_a f_h(\boldsymbol{a}^{(0)}, 0) = 0$. The penalized-block KKT inequality is precisely $\|\nabla_\beta f_h(\boldsymbol{a}^{(0)}, 0)\|_\infty \le \lambda$. This proves (5.19). An approximate nuisance-only fit supplies only approximate KKT conditions, which must be reported rather than treated as exact.

For $z \ge 0$, direct integration and the substitution $t = z + u$ give

$$\phi(z) - z\Phi(-z) = \int_z^{\infty} (t - z)\phi(t)\,\mathrm{d}t = \phi(z) \int_0^{\infty} u \exp(-zu - u^2/2)\,\mathrm{d}u.$$

Since $\ell_h(r) - r_+$ is even, this proves (5.20), including $J(0) = 1$. If $z > 0$, substituting $v = zu$ gives

$$J(z) = z^{-2} \int_0^\infty v\mathrm{e}^{-v} \exp\{-v^2/(2z^2)\} \, \mathrm{d}v.$$

For $x \geq 0$, Taylor's formula for e^{-x} after M terms has remainder of sign $(-1)^M$ and magnitude at most $x^M/M!$. Integrating the finite expansion gives

$$z^{-2} \int_0^\infty v\mathrm{e}^{-v} \frac{(-1)^j v^{2j}}{2^j j! z^{2j}} \, \mathrm{d}v = \frac{(-1)^j (2j+1)!}{2^j j! z^{2j+2}} = \frac{(-1)^j (2j+1)!!}{z^{2j+2}}.$$

The same bound on the integrated remainder proves (5.21). The remainder has the sign of the next term, so adjacent truncations bracket the integral in exact arithmetic. The series is asymptotic, not generally convergent for a fixed finite z; a numerical implementation must stop according to a remainder bound before terms grow. Tail underflow, if encountered, must be distinguished from an exact zero of Φ or ℓ_h in the existence and convergence arguments.

F Numerical reproducibility and work accounting

The computational illustrations use the synthetic design in Section 6.1. Each scenario has one generated training sample. The training-selection record contains all 144 fold candidates, 48 aggregate positions, and 12 selected full-training models, including numerical eligibility and grid-boundary indicators. The three endpoint corrections in Section 6.3 and the nine fold-level corrections in Section 6.4 are separate sets of Newton calls.

F.1 Data generation and training folds

The original synthetic generator uses the Mersenne–Twister, Inversion, and Rejection random-number generator (RNG) settings in R. Matrices are filled by columns, with draw order $\boldsymbol{F}, \boldsymbol{v}, \boldsymbol{Z}, \varepsilon, V$. For scenario index j and replicate $r = 1$, the seed rule is $20260929 + 100000j + 100r +$ role, with roles 1 and 2 for training and test samples. The six training samples have event counts 87, 83, 87, 89, 46, and 93 for S1–S6. Population censoring is calibrated by (6.2); sample censoring retains its generated variation.

The three-fold training-selection analysis uses the event-stratified labels stored with the experiment configuration. Both methods share these labels; each complement contains 80 patients and each validation fold 40. Every mean and PCA transformation is learned on its training complement. The aggregate score retains all within-fold ordered pairs and the common zero-prediction reference in (3.18).

Earlier evaluations of the generated 400-patient test samples form part of the development record. The cross-validation analysis in this article uses training samples throughout. Assessment of the complete selected procedure on new independent samples is a separate evaluation.

F.2 Computational work

A FISTA update is an accepted proximal update, and a Newton update is an accepted active-face correction. The training-selection run has the following work counts; initialization and endpoint verification are counted separately.

Operation	Calls	Accepted updates	Gradient evaluations
Nuisance initialization	48	3,531	3,579 plus 48 full-design
FISTA	156	840,530	1,680,940
Active-face refinement	9	9	18

Endpoint verification additionally uses 165 core and 165 independent pairwise gradient evaluations. The nine refinements require nine Hessian constructions and linear solves; all accept full steps without halving. They occur in UA fold candidates, while the twelve selected full-training refits use no Newton updates. The complete run takes 7,709.2 seconds on the recorded Apple M4 system, including preprocessing, initialization, fitting, verification, and input/output. This is a single-run resource measurement.

F.3 Source objects and environment

The reproducibility record retains all aggregate scores, candidate statuses, selected-model summaries, and final coefficient vectors. The identity map specifies the original fields and source hashes. Coefficient nonzero counts use exact comparison with zero. The final standardized objects reconstruct coefficients, the deployment slope, training scores, gradients, and KKT values from the final coefficient vector, including candidates completed by refinement.

The computation uses R 4.5.2 on Apple M4 arm64 with macOS 26.5, Apple Accelerate Basic Linear Algebra Subprograms (BLAS), and Linear Algebra Package (LAPACK) 3.12.1. The original environment includes Python 3.12.15 for independent checks. Package versions and numerical controls are recorded in the accompanying environment and configuration files. Saved R serialized-data (RDS) objects remain the numerical originals; exported tables and figures are linked to those objects. Full trajectories and checkpoints remain in the computational archive. Artwork was generated in R and reformatted by vector-path-preserving operations with PyMuPDF and Ghostscript; plotted coordinates were retained.

Figure S1 presents the accepted-update counterpart of the main-text stationarity trajectories.

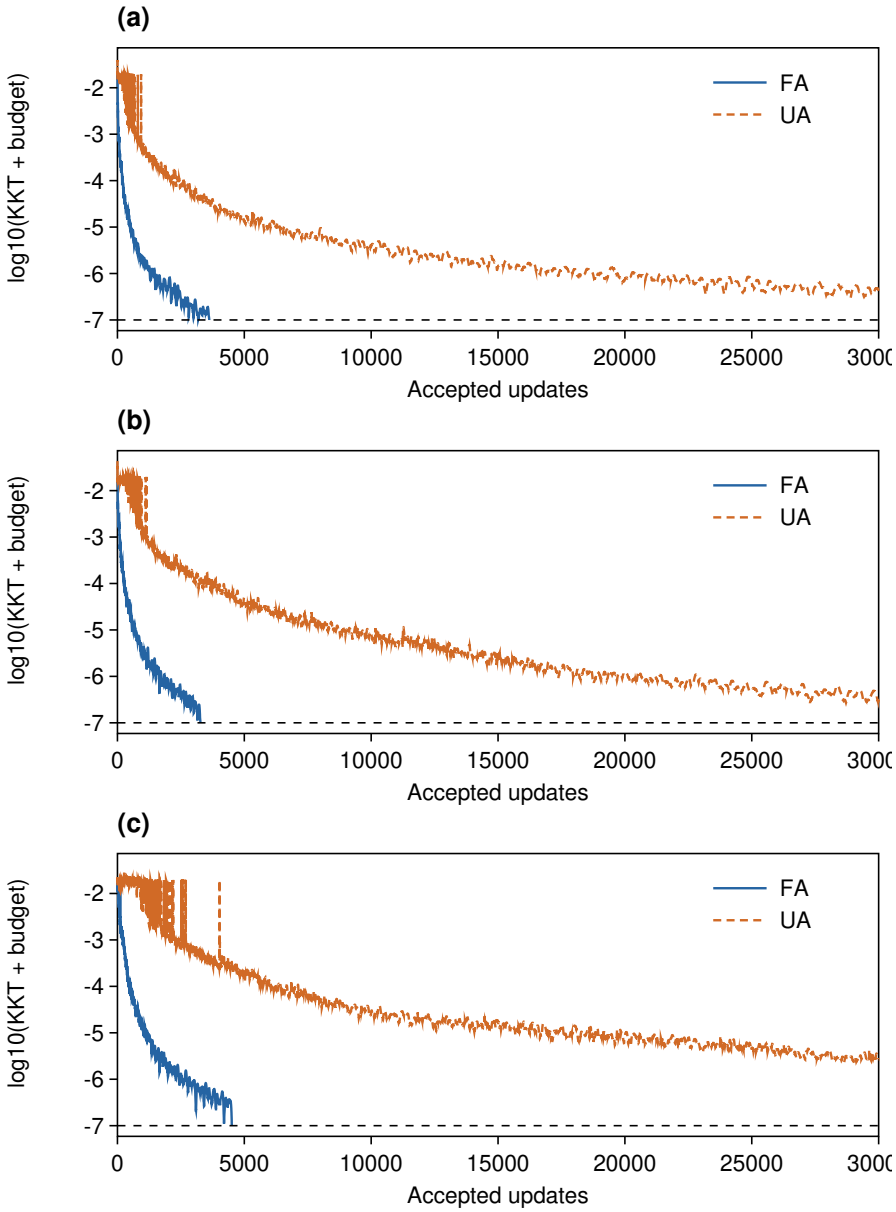


Fig. S1 Full original-coordinate KKT residuals plus tracked numerical budgets versus accepted FISTA updates. The horizontal line is 10^{-7}. Panels (a), (b), and (c) show S3, S4, and S6. FA is solid and UA is dashed. The fits are the same as Figure 1, ending before active-face refinement

G Implementation conventions and numerical diagnostics

Objective and coordinates. The implementation retains ordered off-diagonal pairs, event weights Δ_i, residual direction $e_j - e_i$, and normalization $n(n-1)$. Censored patients remain comparison subjects. Clinical and factor coefficients are unpenalized; molecular coefficients use the declared ℓ_1 penalty and original coordinate scales. For rescaled coordinates, Section 3.1 gives the equivalent weighted penalty. Covariate means and the PCA map are learned within training data and frozen for validation. The recorded observation model determines the treatment of rounded times and ties.

Finite attainment and stationarity. The complete KKT residual uses every nuisance and molecular coordinate and the exact zero/nonzero status of the stored coefficients. The nuisance-direction diagnostic is recorded separately from stationarity. Exact zero of the complete residual characterizes a finite minimizer of the convex criterion. A positive-tolerance numerical result is reported as stationarity to the declared tolerance; the archived output retains `global_optimum_certified=FALSE`.

Tracked arithmetic budget. The numerical allowance E tracks known tail and underflow contributions using normal-tail bounds and log-scale calculations. It is combined with the computed residual for numerical acceptance. It is not a complete interval enclosure of every floating-point operation. Nonfinite predictors, gradients, or losses and unresolved line searches receive distinct status codes. The expansion in (5.21) is truncated using its remainder bound before asymptotic terms grow. A positive quantity below representable range retains its logarithmic bound rather than being interpreted as a mathematical zero.

Stable validation contrasts. Validation computes the positive-part difference within each ordered pair before aggregation. With $a = Y_j - Y_i$ and $b = \widehat{\eta}_i - \widehat{\eta}_j$, this is $(a+b)_+ - a_+$. Sign-based evaluation returns b when both positive parts are active, avoiding cancellation of two large totals. The bound $|(a+b)_+ - a_+| \le |b|$ provides a deterministic arithmetic check. Comparisons are formed within each

validation fold, and the denominator is the sum of its ordered-pair counts.

Paths, refinement, and candidate eligibility. The null-fit threshold is evaluated after nuisance-only fitting and a complete-design gradient calculation, with the numerical budget propagated to its reported status. Penalty levels remain strictly positive. Warm starts advance from the last successful candidate. Active-face refinement retains the input support and signs, directly solves a numerically positive-definite Hessian system, and verifies all inactive-coordinate inequalities at the accepted point. The procedure in Section 5.9 fixes the trigger and update budget. An invalid fold makes the corresponding aggregate unavailable; all candidate positions, grid boundaries, and status reasons remain in the record.

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